United States Patent	12410507
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Nakamura; Yusuke et al.

Mask group, method of manufacturing organic device and organic device

Abstract

A mask group may include two or more masks. When a section in a normal direction is viewed, a region-defining straight line may be defined as a straight line that forms an angle θ together with a first surface. The region-defining straight line may intersect the first surface at a first intersection point. An effective region may be defined as a region inside the first intersection point in the through-hole, and a peripheral region may be defined as a region outside the first intersection point in the through-hole. The angle θ may be 35° or more and 70° or less. The penetration region in the second mask region may include a hole overlapping region in which the through-holes of two masks overlap. The hole overlapping region may include a first hole overlapping region in which the peripheral regions of the through-holes of the two masks overlap.

Inventors: **Nakamura; Yusuke (Tokyo, JP), Miyatani; Isao (Tokyo, JP), Nishimura; Naoki (Tokyo, JP), Yano; Hiroshi (Tokyo, JP)**

Applicant: **Dai Nippon Printing Co., Ltd.** (Tokyo, JP)

Family ID: **83006124**

Assignee: **Dai Nippon Printing Co., Ltd. (Tokyo, JP)**

Appl. No.: **17/806531**

Filed: **June 13, 2022**

Prior Publication Data

Document Identifier	Publication Date
US 20220399503 A1	Dec. 15, 2022

Foreign Application Priority Data

JP	2021-098994	Jun. 14, 2021
JP	2022-037594	Mar. 10, 2022

Publication Classification

Int. Cl.: **C23C14/04** (20060101); **H10K50/82** (20230101); **H10K59/121** (20230101); **H10K59/65** (20230101); **H10K59/80** (20230101); **H10K71/00** (20230101)

U.S. Cl.:

CPC **C23C14/042** (20130101); **H10K50/82** (20230201); **H10K59/121** (20230201); **H10K59/65** (20230201); **H10K59/80521** (20230201); **H10K71/00** (20230201); **H10K71/621** (20230201);

Field of Classification Search

CPC: H01L (21/02636); H01L (21/02639); H01L (21/02642); H01L (21/0268); H01L (21/28562); H01L (21/32139); C23C (16/042); C23C (14/042); C23C (14/24); H10K (50/11); H10K (50/82); H10K (59/121); H10K (59/65); H10K (59/352); H10K (59/353); H10K (59/80521); H10K (71/00); H10K (71/166); H10K (71/621)

USPC: 257/40; 118/504; 427/282; 427/331

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Primary Examiner: Gondarenko; Natalia A

Attorney, Agent or Firm: BURR PATENT LAW, PLLC

Background/Summary

CROSS REFERENCES TO RELATED APPLICATIONS
(1) This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2021-098994 filed on Jun. 14, 2021 and Japanese Patent Application No. 2022-037594 filed on Mar. 10, 2022, the entire contents of which are incorporated herein by reference.
FIELD
(2) An embodiment of the present disclosure relates to a mask group, a method of manufacturing an organic device, and an organic device.
BACKGROUND
(3) For electronic devices such as smart phones and tablet PCs, there has been a need for high-definition display devices in markets. A display device has a pixel density of, for example, 400 ppi or more or 800 ppi or more.
(4) An organic electroluminescent display device attracts attention because of excellent responsiveness and/or low power consumption. A known method of forming pixels of the organic electroluminescent display device is to attach the material of the pixels to a substrate by deposition. For example, a substrate on which an anode is formed in a pattern suitable for an element is first prepared. Subsequently, an organic material is attached to the anode via a through-hole in a mask, and an organic layer is formed on the anode. Subsequently, a conductive material is attached to the

organic layer via the through-hole of the mask, and a cathode is formed on the organic layer.
(5) In some cases, the cathode is formed by using a deposition method in which multiple masks are used. In these cases, layers that are included in the cathode are formed by the masks, and adjacent layers overlap in an electrode overlapping region. Consequently, the layers are electrically connected to each other. Patent Document 1: JP 2019-060028 A Patent Document 2: JP 2005-183153 A
DISCLOSURE
(6) As the thickness of the electrode overlapping region of the cathode increases, optical transmittance decreases.
(7) A mask group according to an embodiment of the present disclosure may include two or more masks. The mask may include a shield region and a through-hole. A mask multilayer body in which the two or more masks are stacked may have a penetration region that overlaps the through-hole when viewed in a normal direction of the masks. The mask multilayer body may have a first mask region that contains the penetration region that has a first aperture ratio and a second mask region that contains the penetration region that has a second aperture ratio lower than the first aperture ratio when viewed in the normal direction of the masks. The masks may have a first surface and a second surface that is located opposite the first surface. The through-hole may include a first recessed portion that is located at the first surface, a second recessed portion that is located at the second surface, and a connection portion at which the first recessed portion and the second recessed portion are connected to each other. When a section in the normal direction is viewed, a region-defining straight line may be defined as a straight line that passes through the connection portion and that forms an angle θ together with the first surface, the region-defining straight line may intersect the first surface at a first intersection point, an effective region may be defined as a region inside the first intersection point in the through-hole, and a peripheral region may be defined as a region outside the first intersection point in the through-hole. The angle θ may be 35° or more and 70° or less. The penetration region in the second mask region may include a hole overlapping region in which the through-holes of the two masks overlap. The hole overlapping region may include a first hole overlapping region in which the peripheral regions of the through-holes of the two masks that are included in the mask multilayer body overlap.
(8) According to an embodiment of the present disclosure, optical transmittance can be increased.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a plan view of an example of an organic device according to an embodiment of the present disclosure.
(2) FIG. **2** is a plan view of a second display region of the organic device.
(3) FIG. **3** is a plan view of second electrode of the organic device.
(4) FIG. **4** is a plan view of the organic device illustrated in FIG. **3** with the second electrode removed therefrom.
(5) FIG. **5** is a sectional view of the organic device illustrated in FIG. **3** substantially taken along line A-A.
(6) FIG. **6** is a sectional view of the organic device illustrated in FIG. **3** substantially taken along line B-B.
(7) FIG. **7** is a plan view of electrode overlapping regions of the second electrode illustrated in FIG. **3**.
(8) FIG. **8** is a sectional view of the electrode overlapping regions illustrated in FIG. **7**.
(9) FIG. **9** is a sectional view of a comparative example of the electrode overlapping regions.
(10) FIG. **10** illustrates an example of a deposition apparatus that includes a mask apparatus.

(11) FIG. **11** is a plan view of an example of the mask apparatus.
(12) FIG. **12** is a plan view of a mask of the mask apparatus.
(13) FIG. **13**A is a plan view of a first mask apparatus.
(14) FIG. **13**B is a plan view of a second mask apparatus.
(15) FIG. **13**C illustrates a third mask apparatus.
(16) FIG. **14** is a sectional view of an example of a mask sectional structure.
(17) FIG. **15** is a sectional view of an effective region and a peripheral region of a through-hole.
(18) FIG. **16** is a plan view of an example of a first mask for forming the second electrode illustrated in FIG. **3**.
(19) FIG. **17** is a plan view of an example of a second mask for forming the second electrode illustrated in FIG. **3**.
(20) FIG. **18** is a plan view of an example of a third mask for forming the second electrode illustrated in FIG. **3**.
(21) FIG. **19** is a plan view of an example of a mask multilayer body for forming the second electrode illustrated in FIG. **3**.
(22) FIG. **20** is a plan view of hole overlapping regions of the mask multilayer body illustrated in FIG. **19**.
(23) FIG. **21** is a sectional view of the hole overlapping regions illustrated in FIG. **20**.
(24) FIG. **22** is a plan view of a modification to the second electrode illustrated in FIG. **3**.
(25) FIG. **23** is a plan view of the electrode overlapping regions of the second electrode illustrated in FIG. **22**.
(26) FIG. **24** is a sectional view of the electrode overlapping regions illustrated in FIG. **23**.
(27) FIG. **25** is a plan view of the hole overlapping regions of the mask multilayer body for forming the second electrode illustrated in FIG. **22**.
(28) FIG. **26** is a sectional view of the hole overlapping regions illustrated in FIG. **25**.
(29) FIG. **27** is a sectional view of a modification to the effective region and the peripheral region of the through-hole illustrated in FIG. **15**.
(30) FIG. **28** is a plan view of a modification to the second electrode illustrated in FIG. **3**.
(31) FIG. **29** is a plan view of the electrode overlapping regions of the second electrode illustrated in FIG. **28**.
(32) FIG. **30** is a plan view of an example of the first mask for forming the second electrode illustrated in FIG. **28**.
(33) FIG. **31** is a plan view of an example of the second mask for forming the second electrode illustrated in FIG. **28**.
(34) FIG. **32** is a plan view of the hole overlapping regions of the mask multilayer body for forming the second electrode illustrated in FIG. **28**.
(35) FIG. **33** is a plan view of the hole overlapping regions of the mask multilayer body illustrated in FIG. **32**.

DETAILED DESCRIPTION

(36) In the present specification and the drawings, words that mean substances on which components such as a “substrate”, a “base material”, a “plate”, a “sheet”, and a “film” are based are not distinguished from each other only based on different names unless there is a particular description.
(37) In the present specification and the drawings, words such as “parallel” and “perpendicular” and the values of a length and an angle that specify shapes, geometrical conditions, and the degree of these, for example, are not limited by strict meaning but are interpreted to an extent that the same function can be expected unless there is a particular description.
(38) In some cases where a component such as a member or a region is “on” or “under”, “on an upper side of” or “on a lower side of”, or “above” or “below” another component such as another member or another region in the present specification and the drawings, the cases include a case

where the component is in direct contact with the other component unless there is a particular description. The cases also include a case where another component is between the component and the other component, that is, a case where the component and the other component are indirectly connected to each other. As for the words “on”, “upper side”, and “above”, or “under”, “lower side”, and “below”, an up-down direction may be reversed unless there is a particular description.
(39) In the present specification and the drawings, like portions or portions that have the same function are designated by like reference characters or similar reference characters unless there is a particular description, and a duplicated description for these is omitted in some cases. For convenience of description, the ratio of dimensions in the drawings differs from an actual ratio, and an illustration of a part of a component is omitted in the drawings in some cases.
(40) In the present specification and the drawings, an embodiment of the present disclosure may be combined with another embodiment or a modification without contradiction unless there is a particular description. Other embodiments may be combined with each other, and another embodiment and a modification may be combined with each other without contradiction. Modifications may be combined with each other without contradiction.
(41) In the case where multiple processes related to a method such as a manufacturing method are disclosed in the present specification and the drawings, another process that is not disclosed may be performed between the disclosed processes unless there is a particular description. The order of the disclosed processes is freely selected without contradiction.
(42) In the present specification and the drawings, a range that is represented by the character “to” includes numerals or factors placed in front of and behind the character “to” unless there is a particular description. For example, the numerical range that is defined by the expression “34 to 38 mass %” is the same as the numerical range that is defined by the expression “34 mass % or more and 38 mass % or less”. For example, the range that is defined by the expression “masks **50**A to **50**C” includes masks **50**A, **50**B, and **50**C.
(43) According to an embodiment of the present specification, an example of the use of a mask group that includes multiple masks to form an electrode on a substrate when an organic electroluminescent display device is manufactured will be described. However, the use of the mask group is not particularly limited. The present embodiment can be used for mask groups that are used in various ways. For example, the mask group according to the present embodiment may be used to form an electrode of a device that displays or projects an image or a picture for creating virtual reality, that is, VR or augmented reality, that is, AR. The mask group according to the present embodiment may be used to form an electrode of a display device other than an organic electroluminescent display device such as an electrode of a liquid-crystal display device. The mask group according to the present embodiment may be used to form an electrode of an organic device other than a display device such as an electrode of a pressure sensor.
(44) According to a first aspect of the present disclosure, a mask group includes two or more masks. The mask includes a shield region and a through-hole. A mask multilayer body in which the two or more masks are stacked has a penetration region that overlaps the through-hole when viewed in a normal direction of the mask. The mask multilayer body has a first mask region that contains the penetration region that has a first aperture ratio and a second mask region that contains the penetration region that has a second aperture ratio lower than the first aperture ratio when viewed in the normal direction of the mask. The mask has a first surface and a second surface that is located opposite the first surface. The through-hole includes a first recessed portion that is located at the first surface, a second recessed portion that is located at the second surface, and a connection portion at which the first recessed portion and the second recessed portion are connected to each other. When a section in the normal direction is viewed, a region-defining straight line is defined as a straight line that passes through the connection portion and that forms an angle θ together with the first surface, the region-defining straight line intersects the first surface at a first intersection point, an effective region is defined as a region inside the first intersection

point in the through-hole, and a peripheral region is defined as a region outside the first intersection point in the through-hole. The angle θ is 35° or more and 70° or less. The penetration region in the second mask region includes a hole overlapping region in which the through-holes of the two masks overlap. The hole overlapping region includes a first hole overlapping region in which the peripheral regions of the through-holes of the two masks that are included in the mask multilayer body overlap.
(45) According to a second aspect of the present disclosure, in the mask group according to the first aspect described above, the first hole overlapping region may be separated from the effective region of the through-holes of the mask.
(46) According to a third aspect of the present disclosure, in the mask group according to the first aspect described above, the hole overlapping region may include a second hole overlapping region in which the effective region of the through-hole of one of the masks that are included in the mask multilayer body and the peripheral region of the through-hole of another of the masks overlap.
(47) According to a fourth aspect of the present disclosure, in the mask group according to any one of the first aspect described above to the third aspect described above, the region-defining straight line may be in contact with a freely selected point on a wall surface of the second recessed portion.
(48) According to a fifth aspect of the present disclosure, in the mask group according to any one of the first aspect described above to the fourth aspect described above, the two or more masks may include a plurality of the through-holes in the first mask region and the second mask region.
(49) According the sixth aspect of the present disclosure, a method of manufacturing an organic device may include a second electrode formation step of forming a second electrode on an organic layer on a first electrode on a substrate by using the mask group according to any one of the first aspect described above to the fifth aspect described above. The second electrode formation step may include forming a first layer of the second electrode by using a deposition method in which one of the masks is used and forming a second layer of the second electrode by using a deposition method in which another of the masks is used.
(50) According to a seventh aspect of the present disclosure, in the method according to the sixth aspect described above, an angle θ of the region-defining straight line may be equal to an angle θ**1**. θ**1** may be an angle that is formed between the first surfaces of the mask and a coming direction in which a deposition material for forming the second electrode comes. θ**2** may be an angle that is formed between the first surfaces and a straight line that passes through the connection portion and that is in contact with the second recessed portion. The angle θ**1** is larger than the angle θ**2**.
(51) According to an eighth aspect of the present disclosure, an organic device includes a substrate, a first electrode that is located on the substrate, organic layers that are located on the first electrode, and a second electrode that is located on the organic layers. The organic device has a first display region that contains the second electrode that has a first occupancy ratio and a second display region that contains the second electrode that has a second occupancy ratio lower than the first occupancy ratio when viewed in a normal direction of the substrate. The second electrode includes two or more layers that are located on the organic layers that differ from each other. The layer has a main layer region and a peripheral layer region thinner than the main layer region. The second electrode in the second display region has an electrode overlapping region in which the two layers overlap. The electrode overlapping region includes a first electrode overlapping region in which the peripheral layer regions of the layers overlap and a second electrode overlapping region in which the main layer region of one of the layers and the peripheral layer region of another of the layers overlap.
(52) An embodiment of the present disclosure will be described in detail with reference to the drawings. The embodiment described below is an example of an embodiment of the present disclosure, and the present disclosure is not interpreted so as to be limited to only the embodiment.
(53) An organic device **100** will be described. The organic device **100** includes an electrode that is formed by using a mask group according to the present embodiment. FIG. **1** is a plan view of an

example of the organic device **100** viewed in the normal direction of a substrate **110** of the organic device **100**. In the following description, viewing a surface of a substance on which the substrate, for example, is based in the normal direction is also referred to as “in a plan view”.

(54) The organic device **100** includes the substrate **110** (see FIG. **5**) and multiple elements **115** that are arranged on a first surface **111** or a second surface **112** of the substrate **110**. For example, the elements **115** are pixels. As illustrated in FIG. **1**, the organic device **100** may have a first display region **101** and a second display region **102** in a plan view. The second display region **102** may have an area smaller than that of the first display region **101**. As illustrated in FIG. **1**, the second display region **102** may be surrounded by the first display region **101**. A part of an outer edge of the second display region **102** may be located on the same straight line as a part of an outer edge of the first display region **101** although this is not illustrated.

(55) FIG. **2** is an enlarged plan view of the second display region **102** and the vicinity thereof in FIG. **1**. In the first display region **101**, the elements **115** may be arranged in two different directions. In an example illustrated in FIG. **1** and FIG. **2**, two or more elements **115** in the first display region **101** may be arranged in a first element direction G**1**. Two or more elements **115** in the first display region **101** may be arranged in a second element direction G**2** that intersects the first element direction G**1**. The second element direction G**2** may be perpendicular to the first element direction G**1**.

(56) The organic device **100** includes a second electrode **140**. The second electrode **140** is located on organic layers **130** described later. The second electrode **140** may be electrically connected to two or more organic layers **130**. For example, the second electrode **140** may overlap the two or more organic layers **130** in a plan view. The second electrode **140** that is located in the first display region **101** is also referred to as second electrode **140**X. The second electrode **140** that is located in the second display region **102** is also referred to as second electrode **140**Y.

(57) The second electrode **140**X has a first occupancy ratio. The first occupancy ratio is calculated by dividing the sum of the areas of the second electrode **140** that is located in the first display region **101** by the area of the first display region **101**. The second electrode **140**Y has a second occupancy ratio. The second occupancy ratio is calculated by dividing the sum of the areas of the second electrode **140** that is located in the second display region **102** by the area of the second display region **102**. The second occupancy ratio may be lower than the first occupancy ratio. For example, as illustrated in FIG. **2**, the second display region **102** may include non-transparent regions **103** and transparent regions **104**. The transparent region **104** do not overlap the second electrode **140**Y in a plan view. The non-transparent region **103** overlap the second electrode **140**Y in a plan view.

(58) The first occupancy ratio may be higher than 0% and lower than 100%. In this case, the first display region **101** includes a region that is occupied by any of first layers **140**A to third layers **140**C of the second electrode **140** described later and a region that is occupied by none of the first layers **140**A to the third layers **140**C. However, the present disclosure is not limited thereto. The first occupancy ratio may be 100%. In this case, the entire first display region **101** is occupied by any of the first layers **140**A to the third layers **140**C of the second electrode **140**. The second occupancy ratio may be higher than 0% and lower than 100%. In this case, the second display region **102** includes a region that is occupied by any of the first layers **140**A to the third layers **140**C of the second electrode **140** and a region that is occupied by none of the first layers **140**A to the third layers **140**C.

(59) For example, the ratio of the second occupancy ratio to the first occupancy ratio may be 0.2 or more, may be 0.3 or more, or may be 0.4 or more. For example, the ratio of the second occupancy ratio to the first occupancy ratio may be 0.6 or less, may be 0.7 or less, or may be 0.8 or less. The range of the ratio of the second occupancy ratio to the first occupancy ratio may be determined by using a first group consisting of 0.2, 0.3, and 0.4 and/or a second group consisting of 0.6, 0.7, and 0.8. The range of the ratio of the second occupancy ratio to the first occupancy ratio may be

determined by using a combination of a value freely selected from the values in the first group described above and a value freely selected from the values in the second group described above. The range of the ratio of the second occupancy ratio to the first occupancy ratio may be determined by using a combination of two values freely selected from the values in the first group described above. The range of the ratio of the second occupancy ratio to the first occupancy ratio may be determined by using a combination of two values freely selected from the values in the second group described above. For example, the range may be 0.2 or more and 0.8 or less, may be 0.2 or more and 0.7 or less, may be 0.2 or more and 0.6 or less, may be 0.2 or more and 0.4 or less, may be 0.2 or more and 0.3 or less, may be 0.3 or more and 0.8 or less, may be 0.3 or more and 0.7 or less, may be 0.3 or more and 0.6 or less, may be 0.3 or more and 0.4 or less, may be 0.4 or more and 0.8 or less, may be 0.4 or more and 0.7 or less, may be 0.4 or more and 0.6 or less, may be 0.6 or more and 0.8 or less, may be 0.6 or more and 0.7 or less, or may be 0.7 or more and 0.8 or less.
(60) The transmittance of the non-transparent region **103** is also referred to as a first transmittance. The transmittance of the transparent region **104** is also referred to as a second transmittance. The transparent region **104** do not contain the second electrode **140**Y, and accordingly, the second transmittance is higher than the first transmittance. For this reason, in the second display region **102** including the transparent regions **104**, light that reaches the organic device **100** passes through the transparent region **104** and can reach, for example, an optical component at the back of the substrate **110**. An example of the optical component is a component that detects light to perform a function such as a camera. The second display region **102** includes the non-transparent regions **103**, and accordingly, a picture can be displayed in the second display region **102** in the case where the elements **115** are pixels. The second display region **102** can thus detect light and display a picture. Examples of the function of the second display region **102** that is performed by detecting light include those of a camera and a sensor such as a fingerprint sensor or a face recognition sensor. The amount of light that is received by the sensor can be increased by increasing the second transmittance of the transparent region **104**. The amount of light that is received by the sensor can be increased by decreasing the second occupancy ratio of the second display region **102**.
(61) In the case where any one of the dimensions of the non-transparent regions **103** in the first element direction G**1** and in the second element direction G**2** and the dimensions of the transparent regions **104** in the first element direction G**1** and in the second element direction G**2** is 1 mm or less, the first transmittance and the second transmittance may be measured by using a microspectrophotometer. OSP-SP200 manufactured by Olympus Corporation or any one of LCF series manufactured by Otsuka Electronics Co., Ltd., may be used as the microspectrophotometer. These microspectrophotometers can measure transmittance in a visible range of 380 nm or more and 780 nm or less. Quartz, borosilicate glass for TFT liquid crystal, or alkali glass for TFT liquid crystal may be used as reference. The result of measurement at 550 nm may be used as the first transmittance and the second transmittance.
(62) In the case where any one of the dimensions of the non-transparent regions **103** in the first element direction G**1** and in the second element direction G**2** and the dimensions of the transparent regions **104** in the first element direction G**1** and in the second element direction G**2** is more than 1 mm, the first transmittance and the second transmittance may be measured by using a spectrophotometer. Ultraviolet visible light spectrophotometer UV-2600i or UV-3600i Plus manufactured by SHIMADZU CORPORATION may be used as the spectrophotometer. A fine luminous flux diaphragm unit is mounted on the spectrophotometer. This enables the transmittance of a region that has a dimension of 1 mm at the maximum to be measured. The atmosphere may be used as reference. The result of measurement at 550 nm may be used as the first transmittance and the second transmittance.
(63) For example, a ratio TR**2**/TR**1** of a second transmittance TR**2** to a first transmittance TR**1** may be 1.2 or more, may be 1.5 or more, or may be 1.8 or more. For example, the TR**2**/TR**1** may be 2 or less, may be 3 or less, or may be 4 or less. The range of the TR**2**/TR**1** may be determined by using

a first group consisting of 1.2, 1.5, and 1.8 and/or a second group consisting of 2, 3, and 4. The range of the TR**2**/TR**1** may be determined by using a combination of a value freely selected from the values in the first group described above and a value freely selected from the values in the second group described above. The range of the TR**2**/TR**1** may be determined by using a combination of two values freely selected from the values in the first group described above. The range of the TR**2**/TR**1** may be determined by using a combination of two values freely selected from the values in the second group described above. For example, the range may be 1.2 or more and 4 or less, may be 1.2 or more and 3 or less, may be 1.2 or more and 2 or less, may be 1.2 or more and 1.8 or less, may be 1.2 or more and 1.5 or less, may be 1.5 or more and 4 or less, may be 1.5 or more and 3 or less, may be 1.5 or more and 2 or less, may be 1.5 or more and 1.8 or less, may be 1.8 or more and 4 or less, may be 1.8 or more and 3 or less, may be 1.8 or more and 2 or less, may be 2 or more and 4 or less, may be 2 or more and 3 or less, or may be 3 or more and 4 or less.

(64) As illustrated in FIG. **2**, the second electrode **140**Y may include two or more electrode lines **140**L that are arranged in the first element direction G**1**. The electrode lines **140**L may extend in the second element direction G**2**. For example, the electrode line **140**L may have first end **140**L**1** and second end **140**L**2** that are connected to the second electrode **140**X in the first display region **101**. The second end **140**L**2** is located opposite the first end **140**L**1** in the second element direction G**2**. In the case where a part of the outer edge of the second display region **102** is located on the same straight line as a part of the outer edge of the first display region **101** although this is not illustrated, only one of the ends of the electrode line **140**L may be connected to the second electrode **140**X.

(65) FIG. **3** is an enlarged plan view of the second electrode **140**X in the first display region **101** and the second electrode **140**Y in the second display region **102**. The second electrode **140**X and the second electrode **140**Y may overlap the organic layers **130** in a plan view. The organic layers **130** are components of the elements **115**.

(66) In the first display region **101**, the organic layers **130** may be arranged at a first pitch P**1** (see FIG. **4**) in the first element direction G**1**. In the second display region **102**, the organic layers **130** may be arranged at a second pitch P**2** in the first element direction G**1**. The second pitch P**2** may be larger than the first pitch P**1**. Consequently, the second occupancy ratio of the second electrode **140**Y decreases. This enables the areas of the transparent region **104** to increase and enables the amount of light that is received by the sensor to increase.

(67) For example, the ratio of the second pitch P**2** to the first pitch P**1** may be 1.1 or more, may be 1.3 or more, or may be 1.5 or more. For example, the ratio of the second pitch P**2** to the first pitch P**1** may be 2.0 or less, may be 3.0 or less, or may be 4.0 or less. The range of the ratio of the second pitch P**2** to the first pitch P**1** may be determined by using a first group consisting of 1.1, 1.3, and 1.5 and/or a second group consisting of 2.0, 3.0, and 4.0. The range of the ratio of the second pitch P**2** to the first pitch P**1** may be determined by using a combination of a value freely selected from the values in the first group described above and a value freely selected from the values in the second group described above. The range of the ratio of the second pitch P**2** to the first pitch P**1** may be determined by using a combination of two values freely selected from the values in the first group described above. The range of the ratio of the second pitch P**2** to the first pitch P**1** may be determined by using a combination of two values freely selected from the values in the second group described above. For example, the range may be 1.1 or more and 4.0 or less, may be 1.1 or more and 3.0 or less, may be 1.1 or more and 2.0 or less, may be 1.1 or more and 1.5 or less, may be 1.1 or more and 1.3 or less, may be 1.3 or more and 4.0 or less, may be 1.3 or more and 3.0 or less, may be 1.3 or more and 2.0 or less, may be 1.3 or more and 1.5 or less, may be 1.5 or more and 4.0 or less, may be 1.5 or more and 3.0 or less, may be 1.5 or more and 2.0 or less, may be 2.0 or more and 4.0 or less, may be 2.0 or more and 3.0 or less, or may be 3.0 or more and 4.0 or less. In the case where the ratio of the second pitch P**2** to the first pitch P**1** is small, a difference between the pixel density of the first display region **101** and the pixel density of the second display region

102 is small. This inhibits a visual difference between the first display region **101** and the second display region **102** from being made.
(68) The electrode line **140**L may overlap the two or more organic layers **130** that are arranged in the second element direction G**2** in a plan view.
(69) Layered structures of the second electrode **140** will be described.
(70) The second electrode **140** may include multiple layers. For example, the second electrode **140** may include the first layers **140**A, the second layers **140**B, and the third layers **140**C. The first layers **140**A, the second layers **140**B, and the third layers **140**C are formed by using a deposition method. More specifically, the first layers **140**A are formed by using the deposition method in which a first mask **50**A is used. The second layers **140**B are formed by using the deposition method in which a second mask **50**B is used. The third layers **140**C are formed by using the deposition method in which a third mask **50**C is used.
(71) As illustrated in FIG. **7** and FIG. **8**, each of the layers **140**A to **140**C of the second electrode **140** has main layer region **141** and peripheral layer region **142**.
(72) The main layer region **141** is located at the centers of the layer in a plan view and are unlikely to be affected by shadow described later. The main layer region **141** is thicker than the peripheral layer region **142**. The main layer region **141** may be relatively flat. The main layer region **141** may be defined by effective region **57** of through-hole **53** described later.
(73) The peripheral layer region **142** is located outside the main layer region **141** and around the main layer region **141**. The peripheral layer region **142** is likely to be affected by the shadow. The peripheral layer region **142** is thinner than the main layer region **141**. The thickness of the peripheral layer region **142** may vary. The peripheral layer region **142** may be defined by peripheral region **58** of the through-hole **53** described later.
(74) The peripheral layer region **142** may have a thickness of 95% or less of the thickness ta of the main layer region **141**. That is, the thickness of the peripheral layer region **142** may change within a range of 95% or less of the thickness ta of the main layer region **141**. For example, the percentage of the thickness to the thickness ta may be 80% or less or 50% or less.
(75) The layers **140**A to **140**C of the second electrode **140** may have a contour that has a substantially polygonal shape in a plan view or a contour that has a substantially circular shape. For example, the layers may have a contour that has a substantially quadrilateral shape, a substantially hexagonal shape, or a substantially octagonal shape. In an example illustrated in FIG. **3**, the layers **140**A to **140**C have a contour that has a substantially regular octagonal shape. The layers **140**A to **140**C may have the same planar contour. Two sides that face each other extend in the first element direction G**1**. Other two sides that face each other extend in the second element direction G**2**. In the case where the layers **140**A to **140**C have a contour that has a substantially polygonal shape, four corners of the contour may be curved.
(76) The first layer **140**A may be connected to the second layer **140**B and the third layer **140**C in the first element direction G**1** and in the second element direction G**2**. The first layer **140**A may overlap the organic layer **130** in a plan view. For example, the first layer **140**A may overlap first organic layer **130**A described later. The first layer **140**A may be located on the organic layer **130** that differ from those on which the second layer **140**B and the third layer **140**C are located.
(77) The second layer **140**B may be connected to the first layer **140**A and the third layer **140**C in the first element direction G**1** and in the second element direction G**2**. The second layer **140**B may overlap the organic layer **130** in a plan view. For example, the second layer **140**B may overlap second organic layer **130**B described later. The second layer **140**B may be located on the organic layer **130** that differ from those on which the first layer **140**A and the third layer **140**C are located.
(78) The third layer **140**C may be connected to the first layer **140**A and the second layer **140**B in the first element direction G**1** and in the second element direction G**2**. The third layer **140**C may overlap the organic layer **130** in a plan view. For example, the third layer **140**C may overlap third organic layer **130**C described later. The third layer **140**C may be located on the organic layer **130**

that differ from those on which the first layer **140**A and the second layer **140**B are located.
(79) In the first display region **101**, the first layers **140**A, the second layers **140**B, and the third layers **140**C may be continuously arranged in the first element direction G**1** and in the second element direction G**2**. In the example illustrated in FIG. **3**, the first layer **140**A, the second layer **140**B, and the third layer **140**C are located on vertices of an equilateral triangle. A single pixel is configured by three layers **140**A, **140**B, and **140**C located above, as well as the first electrode **120** and the organic layer **130**.
(80) In the second display region **102**, the first layers **140**A, the second layers **140**B, and the third layers **140**C may be continuously arranged in the first element direction G**1** and in the second element direction G**2**. In the example illustrated in FIG. **3**, a single pixel is configured by three layers **140**A, **140**B, and **140**C that are located on vertices of an equilateral triangle as in the first display region **101**. The pixels are arranged at an interval in the first element direction G**1**. The transparent region **104** are interposed between the adjacent pixels in the first element direction G**1**. The pixels are arranged at an interval in the second element direction G**2**. The second layer **140**B is interposed between two adjacent pixels in the second element direction G**2**. The first electrode **120** and the organic layer **130** are not associated with the second layer **140**B, and insulating layers **160** are interposed between the second layer **140**B that are not included in the pixels and the substrate **110**. The second layers **140**B that are not included in the pixels form the electrode lines **140**L and ensure the conductivity of the second electrode **140**.
(81) FIG. **4** is a plan view of the organic device **100** illustrated in FIG. **3** with the second electrode **140** removed therefrom. The organic layer **130** may include the first organic layers **130**A, the second organic layers **130**B, and the third organic layers **130**C. For example, the first organic layer **130**A, the second organic layer **130**B, and the third organic layer **130**C are red light-emitting layer, blue light-emitting layer, and green light-emitting layer. In the following description, the words and reference characters of the "organic layer **130**" are used in the case where an organic layer structure that is common to the first organic layer **130**A, the second organic layer **130**B, and the third organic layer **130**C is described. The first organic layer **130**A, the second organic layer **130**B, and the third organic layer **130**C are located on vertices of an equilateral triangle as in the three layers **140**A to **140**C of the second electrode **140**.
(82) The arrangement of the second electrode **140** and the organic layer **130** in a plan view is detected by observing the organic device **100** by using a digital microscope having high magnification. The occupancy ratios described above, an area, a dimension, a gap can be calculated based on the result of detection. In the case where the organic device **100** includes a cover such as a glass cover, the second electrode **140** and the organic layer **130** may be observed after the cover is removed. The cover may be removed by being peeled or broken. A scanning electron microscope may be used instead of the digital microscope.
(83) An example of the layered structure of the organic device **100** will now be described. FIG. **5** is a sectional view of the organic device illustrated in FIG. **3** substantially taken along line A-A. FIG. **6** is a sectional view of the organic device illustrated in FIG. **3** substantially taken along line B-B.
(84) The organic device **100** includes the substrate **110** and the elements **115** that are located on the substrate **110** as described above. The element **115** may include the first electrode **120**, the organic layer **130** that are located on the first electrode **120**, and the second electrode **140** that are located on the organic layer **130** as described above.
(85) The organic device **100** may include the insulating layers **160** each of which is located between two first electrodes **120** adjacent to each other in a plan view. The insulating layer **160** may contain, for example, polyimide. The insulating layer **160** may overlap end portions of the first electrode **120**.
(86) The insulating layer **160** may overlap electrode overlapping region **148** described later in a plan view. For example, the electrode overlapping region **148** may be surrounded by the contours of the insulating layer **160** in a plan view.

(87) The electrode overlapping region **148** contain the multiple layers of the second electrode **140**. For this reason, the electrode overlapping region **148** have a transmittance lower than that of a single layer of the second electrode **140**. In some cases where light that passes through the electrode overlapping region **148** exits from the organic device **100**, the intensity of the light is non-uniform. The insulating layer **160** that overlap the electrode overlapping region **148** inhibit the intensity of the light from being non-uniform.
(88) In each electrode overlapping region **148**, two layers partly overlap in a plan view. In the example illustrated in FIG. **3**, the electrode overlapping region **148** includes a region in which the first layer **140**A and the second layer **140**B overlap, a region in which the first layer **140**A and the third layer **140**C overlap, and a region in which the second layer **140**B and the third layer **140**C overlap in a plan view. A region in which three layers **140**A, **140**B, and **140**C overlap may not be included.
(89) As illustrated in FIG. **7** and FIG. **8**, the electrode overlapping region **148** may include first electrode overlapping region **149** in which the peripheral layer regions **142** of the layers overlap. FIG. **7** is a plan view of the electrode overlapping regions **148**. FIG. **8** is a sectional view of the electrode overlapping regions **148**.
(90) For example, the first electrode overlapping region **149** include a region in which the peripheral layer region **142** of the first layer **140**A and the peripheral layer region **142** of the second layer **140**B overlap, a region in which the peripheral layer region **142** of the first layer **140**A and the peripheral layer region **142** of the third layer **140**C overlap, and a region in which the peripheral layer region **142** of the second layer **140**B and the peripheral layer region **142** of the third layer **140**C overlap. According to the present embodiment, the electrode overlapping region **148** include the first electrode overlapping region **149** in which the peripheral layer regions **142** of the layers overlap. In the electrode overlapping region, the main layer regions **141** of the layers may not overlap. The main layer regions **141** of the layers may not overlap.
(91) As illustrated in FIG. **8**, the peripheral layer regions **142** overlap, and the thickness of the second electrode **140** in the electrode overlapping region **148** can be consequently decreased. For example, in the case where the main layer regions **141** overlap as illustrated in FIG. **9**, the maximum thickness tb of the second electrode **140** in the electrode overlapping region **148** is twice the thickness to of the main layer region **141**. In the case where the peripheral layer regions **142** overlap, however, the thickness tb of the second electrode **140** in the electrode overlapping region **148** is less than twice the thickness to of the main layer region **141**. For this reason, the main layer regions **141** do not overlap, and the thickness tb of the second electrode **140** in the electrode overlapping region **148** can be decreased.
(92) As illustrated in FIG. **8**, the first electrode overlapping region **149** may be separated from the main layer region **141**. Non-overlapping region **150** may be located between the first electrode overlapping region **149** and the main layer region **141**.
(93) For example, the first electrode overlapping region **149** in which the peripheral layer region **142** of the first layer **140**A and the peripheral layer region **142** of the second layer **140**B overlap may be separated from the main layer region **141** of the first layer **140**A. The non-overlapping region **150** may be located between the first electrode overlapping region **149** and the main layer region **141** of the first layer **140**A. The first electrode overlapping region **149** in which the peripheral layer region **142** of the first layer **140**A and the peripheral layer region **142** of the second layer **140**B overlap may be separated from the main layer region **141** of the second layer **140**B. The non-overlapping region **150** may be located between the first electrode overlapping region **149** and the main layer region **141** of the second layer **140**B.
(94) Similarly, the first electrode overlapping region **149** in which the peripheral layer region **142** of the first layer **140**A and the peripheral layer region **142** of the third layer **140**C overlap may be separated from the main layer regions **141** of the layers **140**A and **140**C.
(95) Similarly, the first electrode overlapping region **149** in which the peripheral layer region **142**

of the second layer **140**B and the peripheral layer region **142** of the third layer **140**C overlap may be separated from the main layer regions **141** of the layers **140**B and **140**C.

(96) The organic device **100** may be an active-matrix device. For example, the organic device **100** may include switches although this is not illustrated. The switches are electrically connected to the respective multiple elements **115**. Examples of the switches may include transistors. The switches can control on-off of a voltage or current to the respective elements **115**.

(97) The substrate **110** has the first surface **111** and the second surface **112**. The substrate **110** may be an insulating plate member. The substrate **110** preferably has transparency and allow light to pass therethrough.

(98) In the case where the substrate **110** has predetermined optical transparency, the degree of the transparency of the substrate **110** may be such that light from the organic layers **130** can pass therethrough and can be displayed. For example, the transmittance of the substrate **110** in a visible-light region may be 70% or more or may be 80% or more. The transmittance of the substrate **110** may be measured by using a test method for Plastics-Determination of the total luminous transmittance of transparent materials based on JIS K7361-1.

(99) The substrate **110** may be flexible or may not be flexible. The material of the substrate **110** may be appropriately selected depending on the use of the organic device **100**.

(100) Examples of the material of the substrate **110** may include inflexible rigid materials such as quartz glass, Pyrex (registered trademark) glass, a synthetic quartz plate, and non-alkali glass and flexible materials such as resin film, an optical resin plate, and thin glass. The substrate may have a multilayer body that includes a barrier layer on a surface of the resin film or barrier layers on surfaces of the resin film.

(101) The thickness of the substrate **110** may be appropriately selected depending on, for example, the material used for the substrate **110** or the use of the organic device **100**. For example, the thickness of the substrate **110** may be 0.005 mm or more. The thickness of the substrate **110** may be 5 mm or less.

(102) The elements **115** may perform a function in a manner in which a voltage is applied between the first electrode **120** and the second electrode **140** or a current flows between the first electrode **120** and the second electrode **140**. For example, in the case where the elements **115** are pixels of an organic electroluminescent display device, the elements **115** can emit light for forming a picture.

(103) The first electrode **120** contain a conductive material. For example, the first electrode **120** may contain metal, a conductive metal oxide or another conductive inorganic material. The first electrode **120** may contain a conductive metal oxide that has transparency such as an indium tin oxide.

(104) The material of the first electrodes **120** may be metal such as Au, Cr, Mo, Ag, or Mg, an indium tin oxide abbreviated as ITO, an indium zinc oxide abbreviated as IZO, a zinc oxide, an inorganic oxide such as indium oxide, or a conductive polymer such as polythiophene doped with metal. These conductive materials may be used alone, or two or more kinds of the materials may be combined and used. In the case where two or more kinds of the materials are used, layers composed of the respective materials may be stacked. An alloy containing two or more kinds of the materials may be used. For example, a magnesium alloy such as MgAg may be used.

(105) The organic layer **130** contain an organic material. When the organic layer **130** are energized, the organic layer **130** can perform a function. The phrase “to be energized” means that a voltage is applied to the organic layer **130**, or a current flows through the organic layer **130**. Examples of the organic layer **130** may include light-emitting layers that emit light when being energized and layers the refractive index or the optical transmittance of which changes when being energized. The organic layer **130** may contain an organic semiconductor material.

(106) Multilayer structure that includes the first electrode **120**, the first organic layer **130**A and the second electrode **140** are also referred to as first element **115**A. Multilayer structure that includes the first electrode **120**, the second organic layer **130**B, and the second electrode **140** are also

referred to as second element **115**B. A multilayer structure that includes the first electrode **120**, the third organic layer **130**C, and the second electrode **140** is also referred to as third element **115**C. In the case where the organic device **100** is an organic electroluminescent display device, the first element **115**A, the second element **115**B, and the third element **115**C are sub-pixels. A single pixel may correspond to a combination of the first element **115**A, the second element **115**B, and the third element **115**C.
(107) In the following description, the words and reference characters of the "elements **115**" are used in the case where an element structure that is common to the first element **115**A, the second element **115**B, and the third element **115**C is described.
(108) When a voltage is applied between the first electrode **120** and the second electrode **140**, the organic layer **130** that are located therebetween are driven. In the case where the organic layer **130** are light-emitting layer, light is emitted from the organic layer **130**, and the light exits from the second electrode **140** to the outside or the first electrode **120** to the outside.
(109) In the case where the organic layer **130** includes light-emitting layer that emit light when being energized, the organic layer **130** may include a hole injection layer, a hole transport layer, an electron transport layer, an electron injection layer, and another layer.
(110) For example, in the case where the first electrode **120** are anodes, the organic layer **130** may include hole injection transport layer between the light-emitting layer and the first electrode **120**. The hole injection transport layer may be hole injection layer that have a hole injection function, may be hole transport layer that have a hole transport function, or may have both of the hole injection function and the hole transport function. The hole injection layer and the hole transport layer may be stacked on the hole injection transport layer.
(111) In the case where the second electrode **140** is cathodes, the organic layer **130** may include electron injection transport layer between the light-emitting layer and the second electrode **140**. The electron injection transport layer may be electron injection layer that have an electron injection function, may be electron transport layer that have an electron transport function, or may have both of the electron injection function and the electron transport function. The electron injection layer and the electron transport layer may be stacked on the electron injection transport layer.
(112) The light-emitting layer contain a light-emitting material. The light-emitting layer may contain an additive that improves leveling properties.
(113) A known material may be used as the light-emitting material. Examples of the light-emitting material include a pigment material, a metal complex material, and a polymeric material.
(114) Examples of the pigment material may include a cyclopentadiene derivative, a tetraphenyl butadiene derivative, a triphenylamine derivative, an oxadiazole derivative, a pyrazoloquinoline derivative, a distyrylbenzene derivative, a distyrylarylene derivative, a silole derivative, a thiophene ring compound, a pyridine ring compound, a perinone derivative, a perylene derivative, an oligothiophene derivative, an oxadiazole dimer, and a pyrazoline dimer.
(115) Examples of the metal complex material may include a metal complex that includes central metal such as Al, Zn, or Be, or rare-earth metal such as Tb, Eu, or Dy and a ligand such as an oxadiazole, thiadiazole, phenylpyridine, phenylbenzimidazole, or quinoline structure, such as an aluminum quinolinol complex, a benzoquinolinol beryllium complex, a benzoxazole zinc complex, a benzothiazole zinc complex, an azomethyl zinc complex, a porphyrin zinc complex, and a europium complex.
(116) Examples of the polymeric material may include a poly(p-phenylene vinylene) derivative, a polythiophene derivative, a poly(p-phenylene) derivative, a polysilane derivative, a polyacetylene derivative, a polyvinylcarbazole derivative, a polyfluorene derivative, a polyquinoxaline derivative, and a copolymer thereof.
(117) The light-emitting layers may contain a dopant in order to improve luminous efficacy or change a luminous wavelength. Examples of the dopant may include a perylene derivative, a coumarin derivative, a rubrene derivative, a quinacridone derivative, a squarylium derivative, a

porphyrin derivative, a styryl pigment, a tetracene derivative, a pyrazoline derivative, decacyclene, phenoxazone, a quinoxaline derivative, a carbazole derivative, and a fluorene derivative. Heavy metal ions such as platinum or iridium ions may be contained as the dopant. An organic metal complex that has phosphorescence may be used. A single kind of a dopant may be used, or two or more kinds of dopants may be used.
(118) For example, materials disclosed in [0126] to [0127] in JP 2010-272891 A or in [0128] to [0129] in WO 2012/132126 A may be used as the light-emitting material and the dopant.
(119) The thickness of the light-emitting layer are not particularly limited provided that a luminous function can be performed by enabling an electron and a hole to be recombined.
(120) For example, the thickness of the light-emitting layer may be 1 nm or more and 500 nm or less.
(121) A hole injection transport material that is used for the hole injection transport layer may be a known material. Examples of the hole injection transport material may include a triazole derivative, an oxadiazole derivative, an imidazole derivative, a polyaryl alkane derivative, a pyrazoline derivative, a pyrazolone derivative, a phenylenediamine derivative, an arylamine derivative, an amino-substituted chalcone derivative, an oxazole derivative, a styrylanthracene derivative, a fluorenone derivative, a hydrazone derivative, a stilbene derivative, a silazane derivative, a polythiophene derivative, a polyaniline derivative, a polypyrrole derivative, a phenylamine derivative, an anthracene derivative, a carbazole derivative, a fluorene derivative, a distyrylbenzene derivative, a polyphenylene vinylene derivative, a porphyrin derivative, and a styrylamine derivative. Other examples of the material can include a Spiro compound, a phthalocyanine compound, or a metal oxide. Compounds disclosed in Japanese Unexamined Patent Application Publication No. 2011-119681, International Publication No. 2012/018082, Japanese Unexamined Patent Application Publication No. 2012-069963, or [0132] in International Publication No. 2012/132126, for example, are appropriately selected and used as the hole injection transport material.
(122) In the case where the hole injection transport layer have a structure in which a hole injection layer and a hole transport layer are stacked, the hole injection layer may contain an additive A, the hole transport layer may contain the additive A, or the hole injection layer and the hole transport layer may contain the additive A. The additive A may be a small molecule compound or a polymer compound. Examples of the additive A may include a fluorine compound, an ester compound, or a hydrocarbon compound.
(123) An electron injection transport material that is used for the electron injection transport layer may be a known material. Examples of the electron injection transport material may include alkali metals, an alkali metal alloy, a halide of alkali metal, alkaline-earth metals, halides of alkaline-earth metals, an alkaline-earth metal oxide, an organic complex of alkali metal, a magnesium halide or oxide, and an aluminum oxide. Examples of the electron injection transport material may include bathocuproine, bathophenanthroline, a phenanthroline derivative, a triazole derivative, an oxadiazole derivative, a pyridine derivative, a nitro-substituted fluorene derivative, an anthraquinonedimethane derivative, a diphenylquinone derivative, a thiopyrandioxide derivative, aromacyclic tetracarboxylic anhydride such as naphthalene or perylene, carbodiimide, a fluorenylidenemethane derivative, an anthraquinonedimethane derivative, an anthrone derivative, a quinoxaline derivative, a metal complex such as a quinolinol complex, a phthalocyanine compound, and a distyrylpyrazine derivative.
(124) Metal doped layer may be formed by doping an electron transport organic material with alkali metal or alkaline-earth metal and may be used as the electron injection transport layer. Examples of the electron transport organic material include metal complexes such as bathocuproine, bathophenanthroline, a phenanthroline derivative, a triazole derivative, an oxadiazole derivative, a pyridine derivative, and tris(8-quinolinolato) aluminum (Alq.sub.3) and a polymer derivative thereof. Examples of the doped metal may include Li, Cs, Ba, and Sr.

(125) The second electrode **140** contains a conductive material such as metal. The second electrode **140** may be formed on the organic layer **130** by using the deposition method in which the masks are used as described later. Examples of the material of the second electrode **140** may include platinum, gold, silver, copper, iron, tin, chromium, aluminum, indium, lithium, sodium, potassium, calcium, magnesium, chromium, or carbon. These materials may be used alone, or two or more kinds of the materials may be combined and used. In the case where two or more kinds of the materials are used, the second electrode **140** may have a structure in which layers composed of the respective materials are stacked. An alloy containing two or more kinds of the materials may be used for the second electrode **140**. For example, a magnesium alloy such as MgAg, an aluminum alloy such as AlLi, AlCs, or AlMg, an alkali metal alloy, or an alkaline-earth metal alloy may be used for the second electrode **140**.
(126) For example, the thickness ta of the main layer region **141** of the second electrode **140** may be 5 nm or more, may be 10 nm or more, may be 50 nm or more, or may be 100 nm or more. For example, the thickness ta of the main layer region **141** may be 200 nm or less, may be 500 nm or less, may be 1 μm or less, or may be 100 μm or less. The range of the thickness ta of the main layer region **141** may be determined by using a first group consisting of 5 nm, 10 nm, 50 nm, and 100 nm and/or a second group consisting of 200 nm, 500 nm, 1 μm and 100 μm. The range of the thickness ta of the main layer region **141** may be determined by using a combination of a value freely selected from the values in the first group described above and a value freely selected from the values in the second group described above. The range of the thickness ta of the main layer region **141** may be determined by using a combination of two values freely selected from the values in the first group described above. The range of the thickness ta of the main layer region **141** may be determined by using a combination of two values freely selected from the values in the second group described above. For example, the range may be 5 nm or more and 100 μm or less, may be 5 nm or more and 1 μm or less, may be 5 nm or more and 500 nm or less, may be 5 nm or more and 200 nm or less, may be 5 nm or more and 100 nm or less, may be 5 nm or more and 50 nm or less, may be 5 nm or more and 10 nm or less, may be 10 nm or more and 100 μm or less, may be 10 nm or more and 1 μm or less, may be 10 nm or more and 500 nm or less, may be 10 nm or more and 200 nm or less, may be 10 nm or more and 100 nm or less, may be 10 nm or more and 50 nm or less, may be 50 nm or more and 100 μm or less, may be 50 nm or more and 1 μm or less, may be 50 nm or more and 500 nm or less, may be 50 nm or more and 200 nm or less, may be 50 nm or more and 100 nm or less, may be 100 nm or more and 100 μm or less, may be 100 nm or more and 1 μm or less, may be 100 nm or more and 500 nm or less, may be 100 nm or more and 200 nm or less, may be 200 nm or more and 100 μm or less, may be 200 nm or more and 1 μm or less, may be 200 nm or more and 500 nm or less, may be 500 nm or more and 100 μm or less, may be 500 nm or more and 1 μm or less, or may be 1 μm or more and 100 μm or less.
(127) As the thickness to of the main layer region **141** decrease, the transmittance of the second electrode **140** increases, and the transmittance of the non-transparent region **103** increases. Light that enters the non-transparent region **103** can reach the sensor depending on the transmittance of the non-transparent region **103**. The amount of light that is received by the sensor can be increased by increasing the transmittance of the non-transparent region **103**.
(128) The thickness of each component of the organic device **100** may be measured by observing an image of a section of the organic device **100** by using a scanning electron microscope or a white light interferometer. For example, the thickness of the substrate **110** and the thickness of the second electrode **140** may be measured by using a scanning electron microscope or a white light interferometer. For example, the thickness of the main layer region **141** of the second electrode **140** may be measured by using a white light interferometer "VertScan (registered trademark), R6500H-A300" manufactured by Mitsubishi Chemical Systems, Inc.
(129) A method of forming the second electrode **140** of the organic device **100** described above by using the deposition method will be described. FIG. **10** illustrates a deposition apparatus **10**. The

deposition apparatus **10** performs a deposition process for depositing a deposition material on a subject.

(130) The deposition apparatus **10** may contain a deposition source **6**, a heater **8**, and a mask apparatus **40**. The deposition apparatus **10** may include an exhaust unit for obtaining a vacuum atmosphere in the deposition apparatus **10**. An example of the deposition source **6** is a crucible. The deposition source **6** contains a deposition material **7** such as a conductive material. The heater **8** heats the deposition source **6** and vaporizes the deposition material **7** under the vacuum atmosphere. The mask apparatus **40** faces the crucible **6**.

(131) As illustrated in FIG. **10**, the mask apparatus **40** may include at least a single mask **50** and a frame **41** that supports the mask **50**. The frame **41** may have a first frame surface **41***a* and a second frame surface **41***b*. The mask **50** may be fixed to the first frame surface **41***a*. The second frame surface **41***b* is located opposite the first frame surface **41***a*. The frame **41** may have an opening **42**. The opening **42** extends through the frame **41** from the first frame surface **41***a* to the second frame surface **41***b*. The mask **50** may be fixed to the frame **41** so as to be across the opening **42** in a plan view. The frame **41** may support the mask **50** with the mask **50** pulled in the direction of a surface thereof. This inhibits the mask **50** from being bent.

(132) As for the mask **50**, the first mask **50**A, the second mask **50**B or the third mask **50**C described later may be used. In the following description, the words and reference characters of the "mask **50**" are used in the case where a mask structure that is common to the first mask **50**A, the second mask **50**B, and the third mask **50**C is described. In this case, reference characters that contain no alphabet and that contain only numerals such as numerals "53" or "54" are used as in the parts of the mask such as the through-holes and a shield region described later. In some cases where contents that are peculiar to the first mask **50**A, the second mask **50**B, and the third mask **50**C are described, reference characters that contain a suitable alphabet such as "A", "B", or "C" behind a numeral are used.

(133) The mask **50** of the mask apparatus **40** faces the first surface **111** of the substrate **110**. The deposition material **7** is attached to the substrate **110** by using the mask **50**. The mask **50** has the multiple through-holes **53**. The through-holes **53** allow the deposition material **7** that comes from the deposition source **6** to pass therethrough. The deposition material **7** that passes through the through-holes **53** is attached to the first surface **111** of the substrate **110**. The mask **50** have a first surface **51***a* and a second surface **51***b*. The first surface **51***a* faces the first surface **111**. The second surface **51***b* is located opposite the first surface **51***a*. The through-holes **53** extend through the mask **50** from the first surface **51***a* to the second surface **51***b*.

(134) The deposition apparatus **10** may include a substrate holder **2** that holds the substrate **110**. The substrate holder **2** may be movable in the thickness direction of the substrate **110**. The substrate holder **2** may be movable in a direction in which the first surface **111** of the substrate **110** extends. The substrate holder **2** may control the inclination of the substrate **110**. For example, the substrate holder **2** may include multiple chucks that are mounted on outer edges of the substrate **110**. The chucks may be separately movable in the thickness direction of the substrate **110** and in the direction in which the first surface **111** extends.

(135) The deposition apparatus **10** may include a mask holder **3** that holds the mask apparatus **40**. The mask holder **3** may be movable in the thickness direction of the mask **50**. The mask holder **3** may be movable in a direction in which the first surface **51***a* of the mask **50** extends. For example, the mask holder **3** may include multiple chucks that are mounted on outer edges of the frame **41**. The chucks may be separately movable in the thickness direction of the mask **50** and in the direction in which the first surface **51***a* extends.

(136) The position of the mask **50** of the mask apparatus **40** with respect to the substrate **110** can be adjusted by moving at least the substrate holder **2** or the mask holder **3**.

(137) The deposition apparatus **10** may include a cooling plate **4**. The cooling plate **4** may face the second surface **112** of the substrate **110**. The cooling plate **4** may contain a flow path through which

refrigerant circulates in the cooling plate **4**. The cooling plate **4** can inhibit the temperature of the substrate **110** from increasing in a deposition process.

(138) The deposition apparatus **10** may include a magnet **5** facing the second surface **112**. The magnet **5** may be stacked on the cooling plate **4**. The magnet **5** attracts the mask **50** toward the substrate **110** due to magnetic force. This enables a gap between the mask **50** and the substrate **110** to be reduced or eliminated. Consequently, shadow is inhibited from occurring in the deposition process. For this reason, precision of the dimensions of the second electrode **140** and the precision of the positions thereof can be improved. Alternatively, the mask **50** may be attracted toward the substrate **110** by using an electrostatic chuck that uses electrostatic force instead of the magnet **5**.

(139) The mask apparatus **40** will be described. FIG. **11** is a plan view of the mask apparatus **40**. The mask apparatus **40** may include two or more masks **50**. The masks **50** may be fixed to the frame **41** by, for example, welding.

(140) The frame **41** has a pair of first sides **411** and a pair of second sides **412**. The frame **41** may have a rectangular contour. The masks **50** to which a tensile force is applied may be fixed to the first sides **411**. The first sides **411** may be longer than the second sides **412**. The pair of the first sides **411** and the pair of the second sides **412** may surround the opening **42**.

(141) The material of the frame **41** may be the same as the material of the masks **50** described later. An example of the material of the frame **41** may be an iron alloy containing nickel.

(142) Each mask **50** includes at least a cell **52**. The cell **52** includes the through-holes **53** and has a shield region **54** that is located around the through-holes **53**. The cell **52** includes the multiple through-holes **53**. The mask **50** may include two or more cells **52**. In the case where a display device such as an organic electroluminescent display device is manufactured by using the mask **50**, the single cell **52** may correspond to a single display region of the organic electroluminescent display device, that is, a single screen. The single cell **52** may correspond to multiple display regions. The shield region **54** may be located between two cells **52**. The mask **50** may include the through-holes that are located between the two cells **52** although this is not illustrated.

(143) For example, the cell **52** may have a substantially quadrilateral shape in a plan view, more precisely, a substantially rectangular contour in a plan view. The cell **52** may have a contour that has a different shape depending on the shape of the display region of the organic electroluminescent display device. For example, the cell **52** may have a circular contour.

(144) FIG. **12** is an enlarged plan view of an example of the mask **50**. The mask **50** has a first mask direction D**1** and a second mask direction D**2** that intersects the first mask direction D**1**. The first mask direction D**1** may be perpendicular to the second mask direction D**2**. The first mask direction D**1** may coincide with the first element direction G**1**. The second mask direction D**2** may coincide with the second element direction G**2**.

(145) The mask **50** includes the through-holes **53** and the shield region **54** described above. The through-holes **53** are arranged in the first mask direction D**1** and in the second mask direction D**2**.

(146) In the case where the mask **50** is viewed in the normal direction of the first surface **51***a*, the mask **50** has a third mask region M**3** and a fourth mask region M**4**. The third mask region M**3** corresponds to the first display region **101** of the organic device **100** and overlaps a first mask region M**1** of a mask multilayer body **55** described later. The fourth mask region M**4** corresponds to the second display region **102** of the organic device **100** and overlaps a second mask region M**2** of the mask multilayer body **55** described later.

(147) In the third mask region M**3**, the multiple through-holes **53** may be located. In other words, in the first mask region M**1** that overlaps the third mask region M**3**, the multiple through-holes **53** of the mask **50** may be located. The multiple through-holes **53** in the third mask region M**3** may be patterned. For example, the multiple through-holes **53** may be associated with any one of the first layers **140**A to the third layers **140**C of the second electrode **140** in the first display region **101**.

(148) The third mask region M**3** has a third aperture ratio that represents the area ratio of the through-holes **53**. The third aperture ratio is calculated by dividing the sum of the areas of the

through-holes **53** that are located in the third mask region M**3** by the area of the third mask region M**3**. The areas of the through-holes **53** used to calculate the third aperture ratio may be the areas of the through-holes **53** on the first surface **51***a* or the planer area of through-portions **534** described later. The third aperture ratio may be higher than 0% and lower than 100%. In this case, the third mask region M**3** includes a region that is occupied by the through-holes **53** and a region that is not occupied by the through-holes **53**.
(149) In the fourth mask region M**4**, the multiple through-holes **53** may be located. In other words, in the second mask region M**2** that overlaps the fourth mask region M**4**, the multiple through-holes **53** of the mask **50** may be located. The multiple through-holes **53** in the fourth mask region M**4** may be patterned. For example, the multiple through-holes **53** may be associated with any one of the first layers **140**A to the third layers **140**C of the second electrode **140** in the second display region **102**.
(150) The fourth mask region M**4** has a fourth aperture ratio that represents the area ratio of the through-holes **53**. The fourth aperture ratio is calculated by dividing the sum of the areas of the through-holes **53** that are located in the fourth mask region M**4** by the area of the fourth mask region M**4**. The fourth aperture ratio may be lower than the third aperture ratio. The areas of the through-holes **53** used to calculate the fourth aperture ratio may be the areas of the through-holes **53** on the first surface **51***a* or the planer area of the through-portions **534** described later. The fourth aperture ratio may be higher than 0% and lower than 100%. In this case, the fourth mask region M**4** includes a region that is occupied by the through-holes **53** and a region that is not occupied by the through-holes **53**.
(151) For example, the ratio of the fourth aperture ratio to the third aperture ratio may be 0.2 or more, may be 0.3 or more, or may be 0.4 or more. For example, the ratio of the fourth aperture ratio to the third aperture ratio may be 0.6 or less, may be 0.7 or less, or may be 0.8 or less. The range of the ratio of the fourth aperture ratio to the third aperture ratio may be determined by using the first group consisting of 0.2, 0.3, and 0.4 and/or the second group consisting of 0.6, 0.7, and 0.8. The range of the ratio of the fourth aperture ratio to the third aperture ratio may be determined by using a combination of a value freely selected from the values in the first group described above and a value freely selected from the values in the second group described above. The range of the ratio of the fourth aperture ratio to the third aperture ratio may be determined by using a combination of two values freely selected from the values in the first group described above. The range of the ratio of the fourth aperture ratio to the third aperture ratio may be determined by using a combination of two values freely selected from the values in the second group described above. For example, the range may be 0.2 or more and 0.8 or less, may be 0.2 or more and 0.7 or less, may be 0.2 or more and 0.6 or less, may be 0.2 or more and 0.4 or less, may be 0.2 or more and 0.3 or less, may be 0.3 or more and 0.8 or less, may be 0.3 or more and 0.7 or less, may be 0.3 or more and 0.6 or less, may be 0.3 or more and 0.4 or less, may be 0.4 or more and 0.8 or less, may be 0.4 or more and 0.7 or less, may be 0.4 or more and 0.6 or less, may be 0.6 or more and 0.8 or less, may be 0.6 or more and 0.7 or less, or may be 0.7 or more and 0.8 or less.
(152) The mask **50** may have an alignment mark **50**M. For example, the alignment mark **50**M is formed at a corner of the cell **52** of the mask **50**. The alignment mark **50**M may be used for adjusting the position of the mask **50** with respect to the substrate **110** in a process of forming the second electrode **140** on the substrate **110** in the deposition method in which the mask **50** is used. For example, the alignment mark **50**M may be formed at a position at which the alignment mark **50**M overlaps the opening **42** or may be formed at a position at which the alignment mark **50**M overlaps the frame **41**. When the mask apparatus **40** is manufactured, the alignment mark **50**M may be used for adjusting the positions of the mask **50** and the frame **41**.
(153) In the process of forming the second electrode **140**, multiple masks **50** may be used. For example, as illustrated in FIG. **13**A to FIG. **13**C, the multiple masks **50** may include the first mask **50**A, the second mask **50**B, and the third mask **50**C. The first mask **50**A, the second mask **50**B, and

the third mask **50**C may be included in different mask apparatuses **40**. As illustrated in FIG. **13**A, the mask apparatus **40** that includes the first mask **50**A is also referred to as a first mask apparatus **40**A. As illustrated in FIG. **13**B, the mask apparatus **40** that includes the second mask **50**B is also referred to as a second mask apparatus **40**B. As illustrated in FIG. **13**C, the mask apparatus **40** that includes the third mask **50**C is also referred to as a third mask apparatus **40**C.
(154) In the process of forming the second electrode **140**, for example, the first mask apparatus **40**A illustrated in FIG. **13**A is mounted on the deposition apparatus **10**, and the first layers **140**A of the second electrode **140** are formed on the substrate **110**. Subsequently, the second mask apparatus **40**B illustrated in FIG. **13**B is mounted on the deposition apparatus **10**, and the second layers **140**B of the second electrode **140** are formed on the substrate **110**. Subsequently, the third mask apparatus **40**C illustrated in FIG. **13**C is mounted on the deposition apparatus **10**, and the third layers **140**C of the second electrode **140** are formed on the substrate **110**. In this way, in the process of forming the second electrode **140** of the organic device **100**, the multiple masks **50** such as the first mask **50**A, the second mask **50**B, and the third mask **50**C are used in turn. A group of the multiple masks **50** that are used to form the second electrode **140** of the organic device **100** is also referred to as a “mask group”.
(155) FIG. **14** illustrates an example of a sectional structure of the mask **50**. The mask **50** includes a metal plate **51** in which the multiple through-holes **53** are formed. The through-holes **53** extend through the metal plate **51** from the first surface **51***a* to the second surface **51***b*.
(156) The through-holes **53** may include first recessed portions **531** and second recessed portions **532**. The first recessed portions **531** are located at the first surface **51***a*. The second recessed portions **532** are located at the second surface **51***b*. The first recessed portion **531** is connected to the second recessed portion **532** in the thickness direction of the metal plate **51**.
(157) Dimension r**2** of the second recessed portion **532** may be larger than dimension r**1** of the first recessed portion **531** in a plan view. The first recessed portion **531** may be formed by, for example, etching the metal plate **51** from the first surface **51***a*. The second recessed portion **532** may be formed by, for example, etching the metal plate **51** from the second surface **51***b*. The first recessed portion **531** and the second recessed portion **532** are connected to each other at connection portion **533**. The heights h of the connection portion **533** from the first surface **51***a* are also referred to as sectional heights. The sectional heights can affect the shadow described later.
(158) Reference characters **534** represent the through-portion. The areas of the openings of the through-hole **53** in a plan view are smallest at the through-portion **534**. The through-portion **534** may be defined by the connection portion **533**. In FIG. **14**, the through-portion **534** is illustrated by using a dimension r. The dimension r is smaller than the dimension r**1** and smaller than the dimension r**2**.
(159) In the deposition method in which the mask **50** is used, the deposition material **7** passes through the through-portions **534** in the through-holes **53** from the second surface **51***b* to the first surface **51***a*. The deposition material **7** that passes therethrough is attached to the substrate **110**, and the second electrode **140** described above are consequently formed on the substrate **110**. More specifically, the layers such as the first layers **140**A, the second layers **140**B, and the third layers **140**C described above are formed on the substrate **110**. The planar contour of the layer that are formed on the substrate **110** are mainly defined by the planar contour of the through-portion **534**. More specifically, the main layer region **141** of the second electrode **140** are mainly defined by the planar contour of the through-portion **534**. The peripheral layer region **142** of the second electrode **140** are mainly defined by the contours of through-holes **53**A to **53**C on the first surface **51***a*.
(160) In the deposition process for depositing the deposition material **7** on the substrate **110**, a part of the deposition material **7** comes in the normal direction of the substrate **110** from the deposition source **6** toward the substrate **110**. However, a part of the deposition material **7** can come in a direction that inclines with respect to the normal direction. In this case, the part of the deposition material **7** that comes in the inclining direction does not reach the substrate **110** but reaches and

attaches to the wall surface of the through-hole **53** and the second surface **51***b* of the mask **50**. As for the through-hole **53**, the part is likely to be attached to the wall surface of the second recessed portion **532**. For this reason, the thickness of a deposition layer that is formed on the substrate **110** is likely to decrease as the position thereof is nearer to the wall surface of the through-hole **53** although a desired thickness can be maintained at the centers of the through-holes **53**. Such a phenomenon in which attachment of the deposition material **7** to the substrate **110** is impeded by the wall surface of the through-hole **53** and the second surface **51***b* is referred to as the shadow.
(161) The through-hole **53** has the effective region **57** and the peripheral region **58**. The effective region **57** is located at the center of the through-hole **53** in a plan view and is unlikely to be affected by the shadow described above. The peripheral region **58** are located outside the effective region **57** and around the effective region **57**. The peripheral region **58** is likely to be affected by the shadow. The effective region **57** and the peripheral region **58** are located on the first surface **51***a*.
(162) The effective region **57** and the peripheral region **58** may be defined by region-defining straight line L that pass through the connection portion **533** described above. The region-defining straight line L may be defined as a straight line that pass through the connection portion **533** and that form angle θ together with the first surface **51***a* of the mask **50**. The effective region **57** is defined as a region inside the first intersection point CP**1** at which the region-defining straight line L intersects the first surface **51***a* in the through-holes **53**. The peripheral region **58** is defined as a region outside the first intersection point CP**1** in the through-hole **53**.
(163) The region-defining straight line L may be defined by using a direction in which the deposition material **7** comes or sectional shape of the through-hole **53**. In an example illustrated in FIG. **15**, the region-defining straight line L is defined by the sectional shape of the through-hole **53**.
(164) More specifically, as illustrated in FIG. **15**, an angle that is formed between the direction in which the deposition material **7** comes from the deposition source **6** and the first surface **51***a* of the mask **50** is a coming angle θ**1**. Angle that is formed between a straight line that passes through the connection portions **533** and that is in contact with freely selected point on the wall surface of the second recessed portion **532** and the first surface **51***a* of the mask **50** are mask angle θ**2**. In the example illustrated in FIG. **15**, the mask angle θ**2** is larger than the coming angle θ**1**. In this case, the peripheral region **58** that is likely to be affected by the shadow depend on the mask angle θ**2**. For this reason, the angle θ of the region-defining straight line L may be equal to the mask angle θ**2**. In this case, the region-defining straight line L passes through the connection portion **533** and is in contact with the freely selected point on the wall surface of the second recessed portion **532**. In the example illustrated in FIG. **15**, the region-defining straight line L passes through second intersection point CP**2** between the second recessed portion **532** and the second surface **52***b*. In the example illustrated in FIG. **15**, the width of the peripheral region **58** is expressed as a sectional height h/tan θ**2**.
(165) For example, the angle θ**2** that is formed between the region-defining straight line L illustrated in FIG. **15** and the first surface **51***a* may be 35° or more, may be 40° or more, or may be 45° or more. For example, the angle θ**2** may be 50° or less, may be 60° or less, or may be 70° or less. The range of the angle θ**2** may be determined by using a first group consisting of 35°, 40°, and 45° and/or a second group consisting of 50°, 60°, and 70°. The range of the angle θ**2** may be determined by using a combination of a value freely selected from the values in the first group described above and a value freely selected from the values in the second group described above. The range of the angle θ**2** may be determined by using a combination of two values freely selected from the values in the first group described above. The range of the angle θ**2** may be determined by using a combination of two values freely selected from the values in the second group described above. For example, the range may be 35° or more and 70° or less, may be 35° or more and 60° or less, may be 35° or more and 50° or less, may be 35° or more and 45° or less, may be 35° or more and 40° or less, may be 40° or more and 70° or less, may be 40° or more and 60° or less, may be 40° or more and 50° or less, may be 40° or more and 45° or less, may be 45° or more and 70° or

less, may be 45° or more and 60° or less, may be 45° or more and 50° or less, may be 50° or more and 70° or less, may be 50° or more and 60° or less, or may be 60° or more and 70° or less.
(166) The through-holes **53** may have a contour that has a substantially polygonal shape in a plan view or a contour that has a substantially circular shape. For example, the through-holes **53** may have a contour that has a substantially quadrilateral shape, a substantially hexagonal shape, or a substantially octagonal shape.
(167) The shapes of the through-holes **53** may be similar to each other in the thickness direction of the mask **50**. In examples illustrated in FIG. **16** to FIG. **19**, the through-holes **53** have a contour that has a substantially regular octagonal shape. Two sides that face each other extend in the first mask direction D**1**. Other two sides that face each other extend in the second mask direction D**2**. In the case where the through-holes **53** have a contour that has a substantially polygonal shape, four corners of the contour may be curved.
(168) A region of the metal plate **51** other than the through-portions **534** corresponds to the shield region **54** that can shield the deposition material **7** that moves toward the substrate **110** and that is described above.
(169) The shield region **54** in the fourth mask region M**4** may contain a recessed portion that does not extend through the metal plate **51**. The recessed portion that is contained in the fourth mask region M**4** enables the rigidity of the fourth mask region M**4** to be decreased. This enables a difference between the rigidity of the fourth mask region M**4** and the rigidity of the third mask region M**3** to be decreased. This inhibits the mask **50** from having a wrinkle due to the difference in rigidity. For example, the wrinkle is likely to occur when a tensile force is applied to the mask **50**.
(170) For example, the thickness T of the mask **50** may be 5 μm or more, may be 10 μm or more, may be 15 μm or more, or may be 20 μm or more. For example, the thickness T of the mask **50** may be 25 μm or less, may be 30 μm or less, may be 50 μm or less, or may be 100 μm or less. The range of the thickness T of the mask **50** may be determined by using a first group consisting of 5 μm, 10 μm, 15 μm, and 20 μm and/or a second group consisting of 25 μm, 30 μm, 50 μm, and 100 μm. The range of the thickness T of the mask **50** may be determined by using a combination of a value freely selected from the values in the first group described above and a value freely selected from the values in the second group described above. The range of the thickness T of the mask **50** may be determined by using a combination of two values freely selected from the values in the first group described above. The range of the thickness T of the mask **50** may be determined by using a combination of two values freely selected from the values in the second group described above. For example, the range may be 5 μm or more and 100 μm or less, may be 5 μm or more and 50 μm or less, may be 5 μm or more and 30 μm or less, may be 5 μm or more and 25 μm or less, may be 5 μm or more and 20 μm or less, may be 5 μm or more and 15 μm or less, may be 5 μm or more and 10 μm or less, may be 10 μm or more and 100 μm or less, may be 10 μm or more and 50 μm or less, may be 10 μm or more and 30 μm or less, may be 10 μm or more and 25 μm or less, may be 10 μm or more and 20 μm or less, may be 10 μm or more and 15 μm or less, may be 15 μm or more and 100 μm or less, may be 15 μm or more and 50 μm or less, may be 15 μm or more and 30 μm or less, may be 15 μm or more and 25 μm or less, may be 15 μm or more and 20 μm or less, may be 20 μm or more and 100 μm or less, may be 20 μm or more and 50 μm or less, may be 20 μm or more and 30 μm or less, may be 20 μm or more and 25 μm or less, may be 25 μm or more and 100 μm or less, may be 25 μm or more and 50 μm or less, may be 25 μm or more and 30 μm or less, may be 30 μm or more and 100 μm or less, may be 30 μm or more and 50 μm or less, or may be 50 μm or more and 100 μm or less.
(171) A method of measuring the thickness T of the mask **50** may be a contact measurement method. For the contact measurement method, a length gauge HEIDENHAIN-METRO “MT1271” including a ball bush guide plunger, which is manufactured by HEIDENHAIN, may be used.
(172) The sectional shapes of the through-hole **53** are not limited to the shape illustrated in FIG. **14**. A method of forming the through-hole **53** is not limited to etching but may be a different method.

For example, the mask **50** may be formed by plating such that the through-hole **53** is formed.
(173) Examples of the material of the mask **50** can include an iron alloy containing nickel. The iron alloy may contain cobalt in addition to nickel. For example, the material of the mask **50** may be an iron alloy containing 30 mass % or more and 54 mass % or less nickel and cobalt in total and 0 mass % or more and 6 mass % or less cobalt. The iron alloy containing nickel or the iron alloy containing nickel and cobalt may be an invar material containing 34 mass % or more and 38 mass % or less nickel, a super invar material containing cobalt in addition to 30 mass % or more and 34 mass % or less nickel, or a low thermal expansion Fe—Ni plating alloy containing 38 mass % or more and 54 mass % or less nickel. The use of these alloys enables the thermal expansion coefficient of the mask **50** to be decreased. For example, in the case where a glass substrate is used as the substrate **110**, the thermal expansion coefficient of the mask **50** can be decreased to the level of that of the glass substrate. This inhibits the precision of the dimensions of the deposition layer that is formed on the substrate **110** and the precision of the position thereof from decreasing due to the difference in thermal expansion coefficient between the mask **50** and the substrate **110** in the deposition process.
(174) A mask group **56** will be described. The mask group **56** includes two or more masks **50**. According to the present embodiment, the mask group **56** includes the first mask **50**A, the second mask **50**B, and the third mask **50**C described above. A multilayer body obtained by stacking the first mask **50**A, the second mask **50**B, and the third mask **50**C is also referred to as the mask multilayer body **55**.
(175) The first mask **50**A will now be described in detail. FIG. **16** is an enlarged plan view of the third mask region M**3** and the fourth mask region M**4** on the first surface **51***a* of the first mask **50**A. The first mask **50**A includes first through-holes **53**A and a first shield region **54**A. The first through-holes **53**A are arranged in the first mask direction D**1** and in the second mask direction D**2**. In the third mask region M**3** and the fourth mask region M**4**, the first through-holes **53**A are located at suitable positions for the first layers **140**A of the second electrode **140**. The contours of the through-holes **53** illustrated in the plan views in FIG. **16** to FIG. **20** correspond to the contours of the through-holes **53**A to **53**C on the first surfaces **51***a* of the masks **50**A to **50**C. The contours of the through-holes **53**A to **53**C on the first surfaces **51***a* correspond to the contours of the first recessed portions **531** on the first surface **51***a*.
(176) Referring to FIG. **17**, the second mask **50**B will be described. FIG. **17** is an enlarged plan view of the third mask region M**3** and the fourth mask region M**4** on the first surface **51***a* of the second mask **50**B. The second mask **50**B includes second through-holes **53**B and has a second shield region **54**B. The second through-holes **53**B are arranged in the first mask direction D**1** and in the second mask direction D**2** as in the first through-holes **53**A. In the third mask region M**3** and the fourth mask region M**4**, the second through-holes **53**B are located at suitable positions for the second layers **140**B of the second electrode **140**.
(177) Referring to FIG. **18**, the third mask **50**C will be described. FIG. **18** is an enlarged plan view of the third mask region M**3** and the fourth mask region M**4** on the first surface **51***a* of the third mask **50**C. The third mask **50**C includes third through-holes **53**C and a third shield region **54**C. The third through-holes **53**C are arranged in the first mask direction D**1** and in the second mask direction D**2** as in the first through-holes **53**A. In the third mask region M**3** and the fourth mask region M**4**, the third through-holes **53**C are located at suitable positions for the third layers **140**C of the second electrode **140**.
(178) In a method of measuring the shapes and arrangement of the through-holes **53**A to **53**C of the masks **50**A to **50**C, parallel light may be incident on the first surfaces **51***a* or the second surfaces **51***b* in the normal direction of the masks **50**A to **50**C. In this case, the parallel light is emitted from the other of the first surfaces **51***a* or the second surfaces **51***b*. The shape of a region that is occupied by the emitted light may be measured as the shapes of the through-holes **53**. The shape of the region that is occupied by the emitted light may correspond to the shapes of the through-portions

534 described above.
(179) The shapes and arrangement of the through-holes **53**A to **53**C on the first surfaces **51***a* of the masks **50**A to **50**C may be measured by processing images of the first surfaces **51***a*. For example, the first surfaces **51***a* of the masks **50**A to **50**C may be imaged by using a photographing apparatus, and image data about the contours of the through-holes **53**A to **53**C on the first surfaces **51***a* may be obtained.
(180) The positional relationship among the first mask **50**A, the second mask **50**B, and the third mask **50**C will be described. FIG. **19** is a plan view of the mask multilayer body **55**. The mask multilayer body **55** includes the two or more masks **50** that are stacked. The mask multilayer body **55** illustrated in FIG. **19** includes the first mask **50**A, the second mask **50**B, and the third mask **50**C that are stacked.
(181) As for the mask multilayer body **55**, the alignment marks **50**M of the masks **50**A to **50**C may overlap. The masks **50**A to **50**C may be stacked based on the arrangement of the cells **52** of the masks **50**A to **50**C. The masks **50**A to **50**C may be stacked based on the arrangement of the through-holes **53**A to **53**C and the shield regions **54**A to **54**C of the masks **50**A to **50**C. When the masks **50**A to **50**C are stacked, a tensile force may be applied to the masks **50**A to **50**C, or a tensile force may not be applied thereto.
(182) The figure in a state in which the two or more masks **50** are stacked may be obtained by superimposing the image data of the masks **50**. For example, the image data of the first surfaces **51***a* of the masks **50**A to **50**C that is obtained in the above manner by using an image processing apparatus is superimposed. This enables a figure such as FIG. **19** to be obtained. When the image data is obtained, a tensile force may be applied to the masks **50**A to **50**C, or a tensile force may not be applied thereto. The figure in the state in which the two or more masks **50** are stacked may be obtained by superimposing design drawings for manufacturing the masks **50**A to **50**C.
(183) As illustrated in FIG. **19**, the mask multilayer body **55** includes a penetration region **55**A. The penetration region **55**A contains at least one of the through-holes **53**A to **53**C of the masks **50**A to **50**C in a plan view. That is, the penetration region **55**A overlaps at least one of the through-holes **53**A to **53**C of the masks **50**A to **50**C in a plan view. Accordingly, the second electrode **140** that have at least a single layer are formed in a region of the substrate **110** corresponding to the penetration region **55**A in the deposition process. The penetration region **55**A is defined by the contours of the through-holes **53**A to **53**C on the first surfaces **51***a* of the masks **50**A to **50**C.
(184) The mask multilayer body **55** has the first mask region M**1** and the second mask region M**2** in a plan view. The first mask region M**1** corresponds to the first display region **101** of the organic device **100** and overlaps the third mask region M**3** of each mask **50** described above. The second mask region M**2** corresponds to the second display region **102** of the organic device **100** and overlaps the fourth mask region M**4** of each mask **50** described above.
(185) In the first mask region M**1**, the penetration region **55**A has a first aperture ratio. The first aperture ratio represents the area ratio of the penetration region **55**A in the first mask region M**1**. The first aperture ratio is calculated by dividing the total area of the penetration region **55**A that is located in the first mask region M**1** by the area of the first mask region M**1**. The first mask region M**1** may match the third mask region M**3** in a plan view.
(186) The first aperture ratio may be higher than 0% and lower than 100%. In this case, the first mask region M**1** includes a region that is occupied by any of the through-holes **53**A to **53**C and a region that is occupied by none of the through-holes **53**A to **53**C even with the masks **50**A to **50**C stacked. However, the present disclosure is not limited thereto, and the first aperture ratio may be 100%. In this case, the entire first mask region M**1** is occupied by any of the through-holes **53**A to **53**C.
(187) In the second mask region M**2**, the penetration region **55**A has a second aperture ratio. The second aperture ratio represents the area ratio of the penetration region **55**A in the second mask region M**2**. The second aperture ratio is calculated by dividing the total area of the penetration

region **55**A that is located in the second mask region M**2** by the area of the second mask region M**2**. The second mask region M**2** may match the fourth mask region M**4** in a plan view. The second aperture ratio may be lower than the first aperture ratio.

(188) The second aperture ratio may be higher than 0% and lower than 100%. In this case, the second mask region M**2** includes a region that is occupied by any of the through-holes **53**A to **53**C and a region that is occupied by none of the through-holes **53**A to **53**C even with the masks **50**A to **50**C stacked.

(189) For example, the ratio of the second aperture ratio to the first aperture ratio may be 0.2 or more, may be 0.3 or more, or may be 0.4 or more. For example, the ratio of the second aperture ratio to the first aperture ratio may be 0.6 or less, may be 0.7 or less, or may be 0.8 or less. The range of the ratio of the second aperture ratio to the first aperture ratio may be determined by using the first group consisting of 0.2, 0.3, and 0.4 and/or the second group consisting of 0.6, 0.7, and 0.8. The range of the ratio of the second aperture ratio to the first aperture ratio may be determined by using a combination of a value freely selected from the values in the first group described above and a value freely selected from the values in the second group described above. The range of the ratio of the second aperture ratio to the first aperture ratio may be determined by using a combination of two values freely selected from the values in the first group described above. The range of the ratio of the second aperture ratio to the first aperture ratio may be determined by using a combination of two values freely selected from the values in the second group described above. For example, the range may be 0.2 or more and 0.8 or less, may be 0.2 or more and 0.7 or less, may be 0.2 or more and 0.6 or less, may be 0.2 or more and 0.4 or less, may be 0.2 or more and 0.3 or less, may be 0.3 or more and 0.8 or less, may be 0.3 or more and 0.7 or less, may be 0.3 or more and 0.6 or less, may be 0.3 or more and 0.4 or less, may be 0.4 or more and 0.8 or less, may be 0.4 or more and 0.7 or less, may be 0.4 or more and 0.6 or less, may be 0.6 or more and 0.8 or less, may be 0.6 or more and 0.7 or less, or may be 0.7 or more and 0.8 or less.

(190) The penetration region **55**A may include hole overlapping regions **59**. In the hole overlapping regions **59**, the through-holes **53** of the two or more masks **50** overlap in a plan view. That is, the hole overlapping region **59** contains at least two through-holes **53** of the two or more masks **50** that are included in the mask multilayer body **55** in a plan view. More specifically, in the hole overlapping region **59**, the through-holes **53** on the first surfaces **51***a* of the masks **50** overlap.

(191) In an example illustrated in FIG. **19**, the hole overlapping region **59** includes a region in which the first through-hole **53**A and the second through-hole **53**B overlap, a region in which the first through-hole **53**A and the third through-hole **53**C overlap, and a region in which the second through-hole **53**B and the third through-hole **53**C overlap in a plan view. Accordingly, the second electrode **140** that has two layers are formed in region of the substrate **110** corresponding to the hole overlapping region **59** in the deposition process. The penetration region **55**A may not include a region in which the first through-hole **53**A, the second through-hole **53**B, and the third through-hole **53**C overlap.

(192) As illustrated in FIG. **20** and FIG. **21**, the hole overlapping region **59** may include first hole overlapping region **60** in which the peripheral region **58** of the through-holes **53** of two masks **50** that are included in the mask multilayer body **55** overlap. FIG. **20** is a plan view of the hole overlapping regions **59**. FIG. **21** is a sectional view of the hole overlapping regions **59**. In an example in FIG. **21**, the hole overlapping regions **59** that are formed by the first mask **50**A and the second mask **50**B are illustrated.

(193) For example, the first hole overlapping region **60** include a region in which the peripheral region **58** of the first through-hole **53**A and the peripheral region **58** of the second through-hole **53**B overlap, a region in which the peripheral region **58** of the first through-hole **53**A and the peripheral region **58** of the third through-hole **53**C overlap, and a region in which the peripheral region **58** of the second through-hole **53**B and the peripheral region **58** of the third through-hole **53**C overlap. According to the present embodiment, the hole overlapping region **59** include the first

hole overlapping region **60** in which the peripheral region **58** of the through-holes **53** overlap. According to the present embodiment, the effective region **57** of the through-hole **53** do not overlap the hole overlapping region **59**.

(194) The first hole overlapping region **60** may be separated from the effective region **57**. Non-overlapping region **61** may be located between the first hole overlapping region **60** and the effective region **57**.

(195) For example, the first hole overlapping region **60** in which the peripheral region **58** of the first through-hole **53**A and the peripheral region **58** of the second through-hole **53**B overlap may be separated from the effective region **57** of the first through-hole **53**A. The non-overlapping region **61** may be located between the first hole overlapping region **60** and the effective region **57** of the first through-hole **53**A. The first hole overlapping region **60** in which the peripheral region **58** of the first through-hole **53**A and the peripheral region **58** of the second through-hole **53**B overlap may be separated from the effective region **57** of the second through-hole **53**B. The non-overlapping region **61** may be located between the first hole overlapping region **60** and the effective region **57** of the second through-hole **53**B.

(196) Similarly, the first hole overlapping region **60** in which the peripheral region **58** of the first through-hole **53**A and the peripheral region **58** of the third through-hole **53**C overlap may be separated from each of the effective regions **57** of the through-holes **53**A and **53**C. The non-overlapping region **61** may be located between the first hole overlapping region **60** and each of the effective regions **57** of the through-holes **53**A and **53**C.

(197) Similarly, the first hole overlapping region **60** in which the peripheral region **58** of the second through-hole **53**B and the peripheral region **58** of the third through-hole **53**C overlap may be separated from each of the effective regions **57** of the through-holes **53**B and **53**C. The non-overlapping region **61** may be located between the first hole overlapping region **60** and each of the effective regions **57** of the through-holes **53**B and **53**C.

(198) When the penetration region **55**A includes the first hole overlapping regions **60** described above, the second electrode **140** illustrated in FIG. **7** and FIG. **8** can be formed. The first hole overlapping region **60** correspond to the first electrode overlapping region **149** described above.

(199) The area of the hole overlapping region **59** may be smaller than the area of the first through-hole **53**A. For example, the ratio of the area of the hole overlapping region **59** to the area of the first through-hole **53**A may be 0.02 or more, may be 0.05 or more, or may be 0.10 or more. For example, the ratio of the area of the hole overlapping region **59** to the area of the first through-hole **53**A may be 0.20 or less, may be 0.30 or less, or may be 0.40 or less. The range of the ratio of the area of the hole overlapping region **59** to the area of the first through-hole **53**A may be determined by using a first group consisting of 0.02, 0.05, and 0.10 and/or a second group consisting of 0.20, 0.30, and 0.40. The range of the ratio of the area of the hole overlapping region **59** to the area of the first through-hole **53**A may be determined by using a combination of a value freely selected from the values in the first group described above and a value freely selected from the values in the second group described above. The range of the ratio of the area of the hole overlapping region **59** to the area of the first through-hole **53**A may be determined by using a combination of two values freely selected from the values in the first group described above. The range of the ratio of the area of the hole overlapping region **59** to the area of the first through-hole **53**A may be determined by using a combination of two values freely selected from the values in the second group described above. For example, the range may be 0.02 or more and 0.40 or less, may be 0.02 or more and 0.30 or less, may be 0.02 or more and 0.20 or less, may be 0.02 or more and 0.10 or less, may be 0.02 or more and 0.05 or less, may be 0.05 or more and 0.40 or less, may be 0.05 or more and 0.30 or less, may be 0.05 or more and 0.20 or less, may be 0.05 or more and 0.10 or less, may be 0.10 or more and 0.40 or less, may be 0.10 or more and 0.30 or less, may be 0.10 or more and 0.20 or less, may be 0.20 or more and 0.40 or less, may be 0.20 or more and 0.30 or less, or may be 0.30 or more and 0.40 or less.

(200) The area of the hole overlapping region **59** may be smaller than the area of the second through-hole **53**B. The range of the ratio of the area of the hole overlapping region **59** to the area of the second through-hole **53**B may be the range of the "ratio of the area of the hole overlapping region **59** to the area of the first through-hole **53**A" described above.
(201) The area of the hole overlapping region **59** may be smaller than the area of the third through-hole **53**C. The range of the ratio of the area of the hole overlapping region **59** to the area of the third through-hole **53**C may be the range of the "ratio of the area of the hole overlapping region **59** to the area of the first through-hole **53**A" described above.
(202) As illustrated in FIG. **19**, the penetration region **55**A that is located in the second mask region M**2** may include two or more through-lines **55**L that are arranged in the first mask direction D**1**. The through-line **55**L may extend in the second mask direction D**2**. For example, the through-line **55**L may have a third end and a fourth end that are connected to the penetration region **55**A in the first mask region M**1**. The fourth end are located opposite the third end in the second mask direction D**2**.
(203) An example a method of manufacturing the organic device **100** will be described.
(204) The substrate **110** that has the first electrodes **120** are first prepared. For example, the first electrodes **120** are formed in a manner in which a conductive layer for forming the first electrodes **120** is formed on the substrate **110** by, for example, a spattering method, and the conductive layer is subsequently patterned by, for example, a photolithography method. The insulating layer **160** that is located between two first electrodes **120** adjacent to each other in a plan view may be formed on the substrate **110**.
(205) Subsequently, as illustrated in FIG. **5** and FIG. **6**, the organic layers **130** that include the first organic layers **130**A, the second organic layers **130**B, and the third organic layers **130**C are formed on the first electrodes **120**. For example, the first organic layers **130**A may be formed by using the deposition method in which a mask that includes the through-holes suitable for the first organic layers **130**A is used. For example, the first organic layers **130**A can be formed by depositing an organic material on the first electrodes **120** that are associated with the first organic layers **130**A by using the mask. The second organic layers **130**B may be formed by using the deposition method in which a mask that includes the through-holes suitable for the second organic layers **130**B is used. The third organic layers **130**C may be formed by using the deposition method in which a mask that includes the through-holes suitable for the third organic layers **130**C is used.
(206) Subsequently, a process of forming the second electrode may be performed. In the process of forming the second electrode, the second electrode **140** are formed on the organic layers **130** by using the mask group **56** described above. A process of forming the first layers **140**A of the second electrode **140** by using the deposition method in which the first mask **50**A is used may be performed. For example, a conductive material such as metal is deposited on, for example, the organic layers **130** by using the first mask **50**A. This enables the first layers **140**A to be formed. Subsequently, a process of forming the second layers **140**B of the second electrode **140** by using the deposition method in which the second mask **50**B is used may be performed. For example, a conductive material such as metal is deposited on, for example, the organic layers **130** by using the second mask **50**B. This enables the second layers **140**B to be formed. Subsequently, a process of forming the third layers **140**C of the second electrode **140** by using the deposition method in which the third mask **50**C is used may be performed. For example, a conductive material such as metal is deposited on, for example, the organic layers **130** by using the third mask **50**C. This enables the third layers **140**C to be formed. In this way, as illustrated in FIG. **5** and FIG. **6**, the second electrode **140** that include the first layers **140**A, the second layers **140**B, and the third layers **140**C can be formed.
(207) An order in which the first layers **140**A, the second layers **140**B, and the third layers **140**C are formed is not particularly limited. For example, the deposition process may be performed in the order of the third layers **140**C, the second layers **140**B, and the first layers **140**A.

(208) Effect according to the present disclosure will be summarized.
(209) The second electrode **140** in the second display region **102** includes the electrode overlapping region **148** in which the first layer **140**A and the second layer **140**B overlap, and the electrode overlapping region **148** include the first electrode overlapping region **149** in which the peripheral layer region **142** of the first layer **140**A and the peripheral layer region **142** of the second layer **140**B overlap. Consequently, the thickness of the second electrode **140** in the first electrode overlapping region **149** can be decreased, and the transmittance in the first electrode overlapping region **149** can be increased. For this reason, the transmittance in the non-transparent region **103** in the second display region **102** can be increased, and the optical transmittance of the second display region **102** can be increased.
(210) The penetration region **55**A in the second mask region M**2** of the mask multilayer body **55** includes the hole overlapping region **59** in which the through-holes **53** of two masks **50** overlap. The hole overlapping region **59** include the first hole overlapping region **60** in which the peripheral region **58** of the two masks **50** overlap. Consequently, the first electrode overlapping region **149** can be formed such that the thickness of the second electrode **140** can be decreased.
(211) The first electrode overlapping region **149** is separated from the main layer region **141**. Consequently, the main layer region **141** can be inhibited from overlapping the first electrode overlapping region **149**, and the thickness of the second electrode **140** in the first electrode overlapping region **149** can be decreased.
(212) The first hole overlapping region **60** of the mask multilayer body **55** is separated from the effective region **57**. Consequently, the first electrode overlapping region **149** that is separated from the main layer region **141** can be formed.
(213) The masks **50** include the multiple through-holes **53** that are located in the first mask region M**1** and the second mask region M**2**. Consequently, the multiple through-holes **53** can be formed in the third mask regions M**3** and the fourth mask regions M**4** of the masks **50**. For example, the second electrode **140** of the organic device **100** can include the layers **140**A to **140**C that are formed in patterns corresponding to the through-holes **53**A to **53**C of the masks **50**A to **50**C. In this case, the number of the masks **50** for forming the second electrode **140** can be decreased, and the number of times of deposition for forming the second electrode **140** can be decreased. For this reason, the amount of the deposition material **7** used can be decreased, and an environmental load can be decreased. The shield region **54** is formed around the through-holes **53**A to **53**C in the third mask region M**3**, and the material of the masks **50**A to **50**C can be left. For this reason, the mechanical strength of the masks **50**A to **50**C can be maintained.
(214) The embodiment described above can be modified in various ways. Modifications will now be described with reference to the drawings as needed. In the following description and the drawings that are used for the description, a portion that has the same structure as a portion according to the embodiment described above is designated by reference characters like to the reference characters that are used for the portion according to the embodiment described above, and a duplicated description is omitted. In the case where the action and effect according to the embodiment described above are clearly achieved also according to the modifications, the description thereof is omitted in some cases.
(215) A first modification will be described with reference to FIG. **22** to FIG. **26**. FIG. **22** is a plan view of a modification to the second electrode in the second display region. FIG. **23** is a plan view of the electrode overlapping region **148** illustrated in FIG. **22**. FIG. **24** is a sectional view of one of the electrode overlapping regions illustrated in FIG. **23**. FIG. **25** is a plan view of the hole overlapping regions **59**. FIG. **26** is a sectional view of the hole overlapping regions **59** illustrated in FIG. **25**.
(216) For example, as illustrated in FIG. **22** to FIG. **24**, the electrode overlapping region **148** may include the first electrode overlapping region **149** described above and second electrode overlapping regions **151**. The second electrode overlapping region **151** include the main layer

region **141** of one of the layers and the peripheral layer region **142** of another layer.

(217) As illustrated in FIG. **22**, the second electrode **140** may include the first layers **140**A, the second layers **140**B, and the third layers **140**C. In an example illustrated in FIG. **22**, a single first layer **140**A, a single second layer **140**B, and two third layers **140**C are located on the vertices of a rhombus. The two third layers **140**C are located on a diagonal that extends in the first element direction G**1**. The four layers **140**A, **140**B, and **140**C that are arranged as described above and the first electrode **120** and the organic layers **130** described above correspond to a single pixel. In the example illustrated in FIG. **22**, a pixel in the first display region **101** and a pixel in the second display region **102** have the same structure.

(218) As illustrated in FIG. **23** and FIG. **24**, the second electrode overlapping region **151** includes a region in which the main layer region **141** of the first layer **140**A and the peripheral layer region **142** of the second layer **140**B overlap and a region in which the peripheral layer region **142** of the first layer **140**A and the main layer region **141** of the second layer **140**B overlap. For example, the second electrode overlapping region **151** include a region in which the main layer region **141** of the first layer **140**A and the peripheral layer region **142** of the third layer **140**C overlap and a region in which the peripheral layer region **142** of the first layer **140**A and the main layer region **141** of the third layer **140**C overlap. For example, the second electrode overlapping region **151** include a region in which the main layer region **141** of the second layer **140**B and the peripheral layer region **142** of the third layer **140**C overlap and a region in which the peripheral layer region **142** of the second layer **140**B and the main layer region **141** of the third layers **140**C overlap. The main layer regions **141** of the layers **140**A to **140**C do not overlap.

(219) As illustrated in FIG. **24**, the main layer region **141** and the peripheral layer region **142** overlap, and the thickness tb of the second electrode **140** in the electrode overlapping region **148** can be consequently increased. Consequently, the sectional area of the electrode overlapping region **148** can be increased, and the electric resistance of the electrode overlapping region **148** can be decreased. However, in the case where the main layer regions **141** do not overlap, the thickness tb of the second electrode **140** in the electrode overlapping region **148** can be less than twice the thickness to of the main layer region **141**.

(220) The mask group **56** for forming the second electrode **140** illustrated in FIG. **22** to FIG. **24** will be described with reference to FIG. **25** and FIG. **26**. FIG. **26** illustrates the hole overlapping regions **59** that are formed by the first mask **50**A and the second mask **50**B by way of example.

(221) As illustrated in FIG. **25** and FIG. **26**, the hole overlapping region **59** of the mask multilayer body **55** may include the first hole overlapping region **60** described above and second hole overlapping region **62**. The second hole overlapping region **62** include the effective region **57** of the through-hole **53** of one of the mask **50** that are included in the mask multilayer body **55** and the peripheral region **58** of the through-hole **53** of another mask **50**.

(222) For example, the second hole overlapping region **62** includes a region in which the effective region **57** of the first through-hole **53**A and the peripheral region **58** of the second through-hole **53**B overlap and a region in which the peripheral region **58** of the first through-hole **53**A and the effective region **57** of the second through-hole **53**B overlap. For example, the second hole overlapping region **62** includes a region in which the effective a region **57** of the first through-hole **53**A and the peripheral region **58** of the third through-hole **53**C overlap and a region in which the peripheral region **58** of the first through-hole **53**A and the effective region **57** of the third through-hole **53**C overlap. For example, the second hole overlapping region **62** include a region in which the effective region **57** of the second through-hole **53**B and the peripheral region **58** of the third through-hole **53**C overlap and a region in which the peripheral region **58** of the second through-hole **53**B and the effective region **57** of the third through-hole **53**C overlap. The effective region **57** of the masks **50** do not overlap.

(223) The penetration region **55**A include the first hole overlapping regions **60** and the second hole overlapping regions **62** described above, and the electrode overlapping region **148** of the second

electrode **140** illustrated in FIG. **23** and FIG. **24** can be consequently formed. The first hole overlapping region **60** corresponds to the first electrode overlapping region **149** described above. The second hole overlapping region **62** corresponds to the second electrode overlapping region **151** described above.

(224) A second modification will be described with reference to FIG. **27**. FIG. **27** is a sectional view of a modification to the effective region **57** and the peripheral region **58** of the mask **50**.

(225) In an example illustrated in FIG. **27**, the region-defining straight line L are defined by using the coming direction in which the deposition material **7** comes.

(226) More specifically, as illustrated in FIG. **27**, the coming angle θ**1** is larger than the mask angle θ**2**. In this case, the peripheral region **58** that is likely to be affected by the shadow depend on the coming angle θ**1**. For this reason, the angle θ of the region-defining straight line L may be equal to the coming angle θ**1**. In this case, the region-defining straight line L are straight line that passes through the connection portion **533** and that form the angle θ**1** together with the first surface **51***a*. In an example illustrated in FIG. **27**, the width of the peripheral region **58** is expressed as the sectional height h/tan θ**1**.

(227) As for the angle θ of the region-defining straight line L illustrated in FIG. **27**, the same numeral example as the angle θ of the region-defining straight line L illustrated in FIG. **15** may be used.

(228) A third modification will be described with reference to FIG. **28** to FIG. **33**. FIG. **28** is a plan view of a modification to the second electrode. FIG. **29** is a plan view of the electrode overlapping region. FIG. **30** is a plan view of a first mask. FIG. **31** is a plan view of a second mask. FIG. **32** is a plan view of the mask multilayer body. FIG. **33** is a plan view of the hole overlapping regions of the mask multilayer body.

(229) In an example illustrated in FIG. **28**, the second electrode **140** may include the first layers **140**A, the second layers **140**B, and the third layers **140**C. The first layers **140**A and the third layers **140**C are formed by using the deposition method in which a first mask **50**D is used. The second layers **140**B are formed by using the deposition method in which a second mask **50**E is used.

(230) The first layers **140**A and the third layers **140**C have a contour that has a substantially regular octagonal shape. The first layers **140**A and the third layers **140**C may have the same planar contour. The second layers **140**B may have a planar contour that differs from those of the first layers **140**A and the third layers **140**C. The second layers **140**B may have a contour that has a substantially octagonal shape that has a longitudinal direction. The longitudinal direction of the second layers **140**B illustrated in FIG. **28** coincides with the second element direction G**2**. Alternatively, the second layers **140**B may have a contour that has a substantially quadrilateral shape that has a longitudinal direction that coincides with the second element direction G**2**. In this case, four corners of the contour of each second layer **140**B may be chamfered.

(231) In the first display region **101**, the first layers **140**A, the second layers **140**B, and the third layers **140**C may be continuously arranged in the first element direction G**1** and in the second element direction G**2**. In the example illustrated in FIG. **28**, a first layer **140**A, two second layers **140**B, and a third layer **140**C are located on the vertices of a quadrilateral shape. The two second layers **140**B are located on a diagonal. The four layers **140**A, **140**B, **140**B, and **140**C that are arranged as described above, the first electrode **120**, and the organic layer **130** described above correspond to a single pixel.

(232) In the second display region **102**, the first layers **140**A, the second layers **140**B, and the third layers **140**C may be continuously arranged in the first element direction G**1** and in the second element direction G**2**. In the example illustrated in FIG. **28**, each of pixels in the second display region **102** has a structure obtained by removing one of the two second layers **140**B of a pixel in the first display region **101**. The pixels in the second display region **102** are arranged at an interval in the first element direction G**1**, and the transparent regions **104** are interposed between the adjacent pixels in the first element direction G**1**. The pixels in the second display region **102** are

continuously arranged in the second element direction G**2**.

(233) As illustrated in FIG. **29**, the electrode overlapping region **148** of the second electrode **140** may include the first electrode overlapping region **149** in which the peripheral layer region **142** of the layers overlap as in the examples illustrated in FIG. **7** and FIG. **8**. The non-overlapping region **150** may be located between the first electrode overlapping region **149** and the main layer region **141**.

(234) The mask group **56** for forming the second electrode **140** illustrated in FIG. **28** will be described with reference to FIG. **30** to FIG. **33**. FIG. **30** is an enlarged plan view of the third mask region M**3** and the fourth mask region M**4** on the first surface **51***a* of the first mask **50**D. FIG. **31** is an enlarged plan view of the third mask region M**3** and the fourth mask region M**4** on the first surface **51***a* of the second mask **50**E. FIG. **32** is a plan view of the hole overlapping region **59**. FIG. **33** is a sectional view of the hole overlapping region **59**.

(235) The mask group **56** according to the third modification includes the first mask **50**D and the second mask **50**E. According to the third modification, the mask multilayer body **55** is obtained by stacking the first mask **50**D and the second mask **50**E.

(236) As illustrated in FIG. **30**, the first mask **50**D includes the first through-holes **53**A, the third through-holes **53**C, and a first shield region **54**D. The first through-holes **53**A and the third through-holes **53**C are arranged in the first mask direction D**1** and in the second mask direction D**2**. In the third mask region M**3** and the fourth mask region M**4**, the first through-holes **53**A are located at suitable positions for the first layers **140**A of the second electrode **140**, and the third through-holes **53**C are located at suitable positions for the third layers **140**C of the second electrode **140**. The contours of the through-holes **53**A to **53**C illustrated in the plan views in FIG. **30** to FIG. **33** are the contours of the through-holes **53**A to **53**C on the first surfaces **51***a* of the masks **50**D and **50**E. The contours of the through-holes **53**A to **53**C on the first surfaces **51***a* correspond to the contours of the first recessed portions **531** on the first surfaces **51***a*.

(237) As illustrated in FIG. **31**, the second mask **50**E includes the second through-holes **53**B and a second shield region **54**E. The second through-holes **53**B are arranged in the first mask direction D**1** and in the second mask direction D**2**. In the third mask region M**3** and the fourth mask region M**4**, the second through-holes **53**B are located at suitable positions for the second layers **140**B of the second electrode **140**.

(238) The first through-holes **53**A and the third through-holes **53**C have a contour that has a substantially regular octagonal shape. The first through-holes **53**A and the third through-holes **53**C may have the same planar contour. The second through-holes **53**B may have a planar contour that differs from those of the first through-holes **53**A and the third through-holes **53**C. The second through-holes **53**B may have a contour that has a substantially octagonal shape that has a longitudinal direction. The longitudinal direction of the second through-hole **53**B illustrated in FIG. **31** coincides with the second mask direction D**2**. Alternatively, the second through-holes **53**B may have a contour that has a substantially quadrilateral shape that has a longitudinal direction that coincides with the second mask direction D**2**. In this case, four corners of the contour of each second through-hole **53**B may be chamfered.

(239) As illustrated in FIG. **32**, the mask multilayer body **55** has the penetration region **55**A. The penetration region **55**A contains at least one of the through-holes **53**A to **53**C of the masks **50**D and **50**E in a plan view. That is, the penetration region **55**A overlap at least one of the through-holes **53**A to **53**C of the masks **50**D and **50**E in a plan view.

(240) As illustrated in FIG. **32**, the penetration region **55**A may include the hole overlapping regions **59** as in the example illustrated in FIG. **19**. As illustrated in FIG. **33**, the hole overlapping region **59** may include the first hole overlapping region **60** as in the examples illustrated in FIG. **20** and FIG. **21**. The non-overlapping region **61** may be located between the first hole overlapping region **60** and the effective region **57**.

(241) The modifications to the embodiment described above are described. Naturally, the multiple modifications can be appropriately combined and used.

Claims

1. A mask group comprising: two or more masks, wherein each of the two or more masks includes a shield region and a through-hole, wherein a mask multilayer body in which the two or more masks are stacked has a penetration region that overlaps the through-hole when viewed in a normal direction of a mask of the two or more masks, wherein the mask multilayer body has a first mask region that contains the penetration region that has a first aperture ratio and a second mask region that contains the penetration region that has a second aperture ratio lower than the first aperture ratio when viewed in the normal direction of the mask, wherein each mask of the two or more masks has a first surface and a second surface that is located opposite the first surface, wherein the through-hole includes a first recessed portion that is located at the first surface, a second recessed portion that is located at the second surface, and a connection portion at which the first recessed portion and the second recessed portion are connected to each other, wherein when a section in the normal direction is viewed, a region-defining straight line is defined as a straight line that passes through the connection portion and that forms an angle θ together with the first surface, the region-defining straight line intersects the first surface at a first intersection point, an effective region is defined as a region inside the first intersection point in the through-hole, and a peripheral region is defined as a region outside the first intersection point in the through-hole, wherein the angle θ is 35° or more and 70° or less, wherein the penetration region in the second mask region includes a hole overlapping region in which through-holes of two masks of the two or more masks overlap, wherein the hole overlapping region includes a first hole overlapping region in which the peripheral regions of the through-holes of the two masks that are included in the mask multilayer body overlap, and wherein effective regions of the through-holes forming the first hole overlapping region do not overlap.
2. The mask group according to claim 1, wherein the first hole overlapping region is separated from the effective region of the through-holes of the mask.
3. The mask group according to claim 1, wherein the hole overlapping region includes a second hole overlapping region in which the effective region of the through-hole of one of the masks that are included in the mask multilayer body and the peripheral region of the through-hole of another of the masks overlap.
4. The mask group according to claim 1, wherein the region-defining straight line is in contact with a freely selected point on a wall surface of the second recessed portion.
5. The mask group according to claim 1, wherein the two or more masks include a plurality of the through-holes in the first mask region and the second mask region.